Correntropy-Based Spatial-Spectral Robust Sparsity-Regularized Hyperspectral Unmixing

Xiaorun Li, *Member, IEEE*, Risheng Huang, *Student Member, IEEE*, and Liaoying Zhao, *Member, IEEE*

***Abstract*— Hyperspectral unmixing (HU) is a crucial technique for exploiting remotely sensed hyperspectral data, which aims at estimating a set of spectral signatures, called endmembers and their corresponding proportions, called abundances. The performance of HU is often seriously degraded by various kinds of noise existing in hyperspectral images (HSIs). Most of existing robust HU methods are based on the assumption that noise or outlier only exists in one kind of formulation, e.g., band noise or pixel noise. However, in real-world applications, HSIs are unavoidably corrupted by noisy bands and noisy pixels simultaneously, which require robust HU in both the spatial dimension and spectral dimension. Meanwhile, the sparsity of abundances is an inherent property of HSIs and different regions in an HSI may possess various sparsity levels across locations. This article proposes a correntropy-based spatial-spectral robust sparsity-regularized unmixing model to achieve 2-D robustness and adaptive weighted sparsity constraint for abundances simultaneously. The updated rules of the proposed model are efficient to be implemented and carried out by a half-quadratic technique. The experimental results obtained by both synthetic and real hyperspectral data demonstrate the superiority of the proposed method compared to the state-of-the-art methods.**

***Index Terms*— Correntropy, half-quadratic optimization, hyperspectral unmixing (HU), sparsity, spatial-spectral robustness.**

I. Introduction

HYPERSPECTRAL images are collected by hundreds of contiguous narrow spectral band images. Due to the limitation of hyperspectral sensors' spatial resolution and mixing effects of ground materials, a single pixel in the hyperspectral data is commonly mixed by several materials' spectra. Therefore, hyperspectral unmixing (HU) is an important problem for exploiting the wealth of information in hyperspectral data. The task of HU can be divided into two main steps, endmember extraction and abundance estimation. For endmember extraction, there are a number of algorithms have been proposed and widely used to find endmembers, including pixel purity index (PPI) [1], N-FINDR [2], vertex component analysis (VCA) [3], and simplex growing algorithm (SGA) [4]. The relationships among these methods have been investigated in [5]–[7]. Using the found endmembers, the abundances can be estimated by solving the constrained optimization problem under constraints including abundance nonnegativity constraint (ANC), abundance sum-to-one constraint (ASC), and so on [8].

Considering that pure pixels may be absent in the hyperspectral data, a group of algorithms have been developed to regard the unsupervised linear HU as a blind source separation (BSS) problem [9]–[11] obtaining endmembers and abundances simultaneously. Among these methods, nonnegative matrix factorization (NMF) has shown to be attractive in extracting sparse and interpretable representations from a data matrix. NMF aims at decomposing a data matrix into two low-rank matrices with a nonnegative constraint: a basis matrix and a coefficient matrix. The decomposition results provide an intuitive and interpretable representation of data [12]. As an unsupervised method, NMF has been applied to HU and shown its advantages [13], [14]. Unfortunately, the solution space of NMF is quite large if no further constraints are imposed. In addition, due to the nonconvexity of NMF's objection function, the algorithm may be trapped in local minima and is sensitive to noise corruption [15]. Guided by the characteristics of hyperspectral applications, additional constraints have been introduced into NMF to shrink the solution space and lead to better results [9], [16], [17]. Miao and Qi [18] proposed a minimum volume constrained NMF (MVCNMF). Wang *et al.* [19] proposed an endmember dissimilarity constrained NMF method for HU. Sparsity is an important inherent property of hyperspectral data and has been widely imposed as a constraint on abundances for HU [15], [20], [21].

In real applications, hyperspectral data commonly suffer from noise corruption caused by various factors from atmospheric conditions, sensor, and calibration errors, which seriously degrade the unmixing performance of HU methods. A number of robust HU methods have been developed to alleviate the influence of noise [22]–[25]. Ma *et al.* [26] proposed a robust sparse unmixing method adopting $\ell_{2,1}$ norm to achieve robustness for noise and outliers. $\ell_{2,1}$ norm is a rotational invariant norm firstly proposed by Ding *et al.* [27] and has been used to effectively handle noise and outliers in various applications [28]–[31]. In [32], Kong *et al.* proposed a robust formulation of NMF using $\ell_{2,1}$ norm as a loss function ($\ell_{2,1}$-NMF) which accommodated outliers and noise

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in a better way than standard NMF. All the above-mentioned algorithms using $\ell_{2,1}$ norm assume that there exist noise and outliers in the samples (pixels). However, for real-world HSIs, band noise also commonly exists and the signal-to-noise ratio (SNR) varies from band to band. In most real HU experiments, the water vapor absorption bands and some low SNR bands are manually selected and eliminated. This practice requires the prior knowledge of noisy bands and may miss important information contained in these bands [24]. Moreover, the noise intensities of the reserved bands still vary dramatically and should be adaptively considered and utilized. For this consideration, Wang *et al.* [25] proposed a robust HU method with correntropy-based metric (CENMF), which automatically assigns adaptive weights to noisy bands and noise-free bands to suppress the contribution of noisy bands. As a symmetric form of $\ell_{2,1}$ norm, $\ell_{1,2}$ norm is firstly used in the Group Lasso as a sparsity constraint to drive as many feature weights to zero as possible [33]. $\ell_{1,2}$ norm has also been applied to HU to achieve robustness in the spectral dimension. In [24], He *et al.* proposed the sparsity-regularized robust NMF (ℓ_1-RNMF, $\ell_{1/2}$-RNMF) to unmix the hyperspectral data using low-noisy bands and noisy bands simultaneously.

To sum up, a majority of robust NMF methods are usually based on the assumption that there exists only one kind of noise formulation (either pixel noise or band noise). And most of these methods seldom consider the difference of noise intensities among different bands or pixels. Hyperspectral data are assumed to have the same SNR for each band or each pixel in these methods. However, in real-world hyperspectral data, there commonly exist pixel noise and band noise simultaneously and the levels of noise intensity vary [34], [35]. In the spectral dimension, the degraded bands mainly result from atmospheric effects and hyperspectral sensor's properties in different wavelengths. In addition, some not well-registered bands may also interfere with the unmixing performance, particularly for the accuracy of abundances. In the spatial dimension, the signal of pixels is also corrupted by a variety of degradations such as photon noise, dark noise, and fixed-pattern noise [36]. Besides, the noisy pixels are also often corrupted by deadlines and stripes, as well as large digital number (DN) variations caused by the sudden change of the electronic charge in the sensor [24]. To the best of our knowledge, although noisy bands and noisy pixels exist simultaneously in hyperspectral data and degrade the unmixing performance seriously, few of existing unmixing methods consider the two formulations of noise (pixel noise and band noise) simultaneously. For this consideration, we are motivated to propose a spatial-spectral robust unmixing model which takes into consideration band noise and pixel noise jointly so as to achieve robustness in both spatial dimension and spectral dimension.

Furthermore, it can be found that a majority of the existing robust unmixing methods have introduced sparsity constraints to improve unmixing performance, which demonstrates the effectiveness of utilizing the sparsity prior of abundances in an unmixing process. However, most of the existing methods simply impose a sparsity regularizer of the same regularization strength for all the pixels' abundances, e.g., ℓ_1 regularizer in ℓ_1-CENMF [25] and $\ell_{1/2}$ regularizer in $\ell_{1/2}$-RNMF [24]. For real HSIs, the mixing levels of pixels varies significantly across their locations. Imposing a sparseness constraint of the same strength over the entire image may not contribute to the unmixing accuracy of pixels' abundances with different sparsity levels. Enforce sparsity in the regions with high mixing levels may even degrade the unmixing performance. How to incorporate adaptive sparsity constraints into a robust unmixing method would be worthy of study. In this article, we propose a correntropy-based sparsity constraint to solve the above-mentioned problem.

The main contributions of this article can be summarized as two aspects.

1) We investigate the idea of simultaneously considering band noise and pixel noise in the HU model and propose a correntropy-based spatial-spectral robust unmixing model, which achieves robustness to both band noise and pixel noise and improves the unmixing performance effectively for the noisy occasions.
2) We also propose a correntropy-based sparsity constraint to impose structured sparsity constraint adaptive to the sparse levels and further improve the unmixing performance. By incorporating the proposed penalty term, we propose a correntropy-based spatial-spectral robust sparse unmixing model, which is efficiently optimized via alternating minimization through the half-quadratic (HQ) technique. Experiments on synthetic and real data demonstrate the superiority of the proposed method.

The remainder of this article is organized as follows. Section II introduces the related works. Section III proposes the correntropy-based spatial-spectral robust sparsity-regularized (CSsRS) unmixing model and develops the optimization algorithm for the proposed model. Section IV validates the effectiveness of the proposed method using synthetic data and real HSIs. Finally, Section V draws some conclusions for this article.

II. Related Work

A. Nonnegative Matrix Factorization

The classical linear mixing model (LMM) assumes that the observed pixels in HSIs are the linear mixtures of several spectral signatures called endmembers. Consider HSI data $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N] \in \mathbb{R}_+^{L\times N}$, where N is the number of pixels and $_+$ denotes nonnegative value. The formulation of LMM can be represented as

$$\mathbf{X} = \mathbf{WH} + \mathbf{E} \tag{1}$$

where $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_P] \in \mathbb{R}_+^{L\times P}$ denotes the endmember matrix, $\mathbf{H} \in \mathbb{R}_+^{P\times N}$ denotes the relative abundances of endmembers, and $\mathbf{E}$ is the residual term. NMF has received considerable attention and has been widely applied to HU. NMF is designed to find an approximate factorization of $\mathbf{X}$, such that $\mathbf{X} \approx \mathbf{WH}$, where $\mathbf{W} \geq 0$ and $\mathbf{H} \geq 0$. To quantify the quality of the approximate factorization, Euclidean distance is commonly used to measure the distance between $\mathbf{X}$ and $\mathbf{WH}$. The cost function of NMF using Frobenius (i.e., quadratic)

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norm is defined as follows:

$$f(\mathbf{W},\mathbf{H})=\frac{1}{2}\|\mathbf{X}-\mathbf{WH}\|_F^2=\frac{1}{2}\sum_{l=1}^{L}\sum_{n=1}^{N}(\mathbf{X}-\mathbf{WH})_{ln}^2. \quad (2)$$

The multiplication update rules for NMF are presented in [37] to locally minimize the cost function in (2)

$$\mathbf{W}_{lp}\leftarrow\mathbf{W}_{lp}\frac{(\mathbf{XH}^T)_{lp}}{(\mathbf{WHH}^T)_{lp}} \quad (3)$$

$$\mathbf{H}_{pn}\leftarrow\mathbf{H}_{pn}\frac{(\mathbf{W}^T\mathbf{X})_{pn}}{(\mathbf{W}^T\mathbf{WH})_{pn}}. \quad (4)$$

B. Correntropy-Based Nonnegative Matrix Factorization

The main drawback of NMF is that it is easy to fall into local minima and is prone to noise. To address this problem, Wang *et al.* [25] proposed the robust HU method with correntropy-based metric (CENMF), which is essentially a band weighted NMF. The concept of correntropy is proposed in information-theoretic learning (ITL) [38] and proved to be useful in non-Gaussian and nonlinear signal processing [39]. Correntropy is a local similarity measure between two random variables $\mathbf{a}$ and $\mathbf{b}$ defined by

$$V_\sigma(\mathbf{a},\mathbf{b})=E[k_\sigma(\mathbf{a}-\mathbf{b})] \quad (5)$$

where $k_\sigma(\cdot)$ is the Gaussian kernel $k_\sigma(\mathbf{a}-\mathbf{b})=\exp(-(\|\mathbf{a}-\mathbf{b}\|_2^2)/\sigma^2)$ and $E[\cdot]$ denotes the expectation operator. σ is the kernel size (bandwidth) which acts the role of controlling the adjustable observation window. Through the adjustable window, the negative effect of outliers can be effectively eliminated, which is intrinsically different from the conventional techniques of using a threshold to deal with outliers.

Since the joint probability density function of $\mathbf{a}$ and $\mathbf{b}$ is often unknown and only a finite number of data points $\{(\mathbf{a_i},\mathbf{b_i})\}_{i=1}^N$ of $\mathbf{a}$ and $\mathbf{b}$ are available, the sample estimator of correntropy is given by

$$\hat{V}_{N,\sigma}(\mathbf{a},\mathbf{b})=\frac{1}{N}\sum_{i=1}^{N}k_\sigma(\mathbf{a}_i-\mathbf{b}_i). \quad (6)$$

By introducing the correntropy-based metric, the cost function of CENMF model is defined as follows:

$$\begin{aligned}G(\mathbf{W},\mathbf{H})&=\sum_{l=1}^{L}\{-k_{\sigma_1}((\mathbf{X}-\mathbf{WH})_{l,:})\}\\&=\sum_{l=1}^{L}\left\{-\exp\left(-\frac{\|(\mathbf{X}-\mathbf{WH})_{l,:}\|_2^2}{\sigma_1^2}\right)\right\}\end{aligned} \quad (7)$$

where $(\mathbf{X}-\mathbf{WH})_{l,:}$ denotes the lth row of $(\mathbf{X}-\mathbf{WH})$, $k_{\sigma_1}(\mathbf{x})=\exp(-(\|\mathbf{x}\|_2^2)/\sigma_1^2)$, and σ_1 is the kernel size which can be determined following the method in [39] by

$$\sigma_1=\frac{\beta}{2L}\sum_{l=1}^{L}\|(\mathbf{X}-\mathbf{WH})_{l,:}\|_2^2 \quad (8)$$

where β is a constant to control the outlier rejection, which is set to 1 by default following [39] and [25].

TABLE I
EXAMPLES OF PFS φ, WHERE c IS A PARAMETER

Convex PFs	Non-convex PFs
$t^c, 1<c\le 2$	$1-\exp(-ct^2)$
$\sqrt{t^2+c}$	$ct^2/(ct^2+1)$
$\log(\cosh(ct))$	$\log(ct^2+1)$
$\begin{cases}t^2/2 & \|t\|\le c\\ c\|t\|-c^2/2 & \|t\|>c\end{cases}$	$\min\{ct^2,1\}$

It is worth noting that the optimization problem of (7) is subjected to the nonnegative constraints $\mathbf{W}\ge 0$ and $\mathbf{H}\ge 0$, which can be formulated as

$$\min_{\mathbf{W},\mathbf{H}} G(\mathbf{W},\mathbf{H}) \text{ s.t. } \mathbf{W}\ge 0,\ \mathbf{H}\ge 0. \quad (9)$$

HQ technique can be used to solve the minimization problem in (9). According to the theory of convex conjugate [40], [41], we have the following proposition.

Proposition 1: Let φ be a potential function (PF) that satisfies the following assumptions.

H1: $t\to\varphi(t)$ is continuous and increasing on $\mathbb{R}_+$ with $\varphi\not\equiv 0$ and $\varphi(0)=0$,

H2: $t\to\varphi(\sqrt{t})$ is concave on $\mathbb{R}_+$,

H3: $\varphi(t)$ is $\mathcal{C}^1$ on $\mathbb{R}_+$ with $\varphi'(0)=0$,

H4: $\lim_{t\to\infty}\varphi(t)/t^2=0$.

Then, there exists a convex and decreasing function ψ such that

$$\varphi(t)=\inf_{q\ge 0}\left(\frac{1}{2}qt^2+\psi(q)\right) \quad (10)$$

where q is an auxiliary variable and for every fixed $t\ge 0$, the infimum in (10) is reached at $q=\delta(t)$. $\delta(t)$ is given by

$$\delta(t)=\begin{cases}\varphi''(0^+), & \text{if } t=0\\ \varphi'(t)/t, & \text{if } t\ne 0.\end{cases} \quad (11)$$

Many different forms of PFs have been investigated and given in the literatures, e.g., [42]–[44], and [45]. Table I tabulates some commonly used examples of PFs [41]. Noting that Proposition 1 applies to convex and nonconvex functions φ, while in all cases the function ψ is convex and decreasing [42].

Considering the case of $\varphi(t)=1-\exp(-ct^2)$ and $c=1$, we have the following proposition by substituting t with $\|\mathbf{x}\|_2/\sigma$.

Proposition 2: There exists a convex and decreasing function ψ such that

$$-k_\sigma(\mathbf{x})=\min_{q\ge 0}\left(q\frac{\|\mathbf{x}\|_2^2}{/}\sigma^2+\psi(q)\right) \quad (12)$$

and for a fixed $\mathbf{x}$, the minimum is reached at $q=\exp(-(\|\mathbf{x}\|_2^2)/\sigma^2)$.[1]

Then, by utilizing the HQ technique, the following augmented objective function in the enlarged parameter space can

[1]Here, we use q to replace $1/2q$.

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be derived:

$$\hat{G}(\mathbf{W},\mathbf{H},\mathbf{u})=\sum_{l=1}^{L}\left(u_l\frac{\|(\mathbf{X}-\mathbf{WH})_{l,:}\|_2^2}{\sigma_1^2}+\psi_1(u_l)\right) \quad (13)$$

where $\mathbf{u}=[u_1,u_2,\ldots,u_L]^T$ is the auxiliary vector introduced by HQ optimization.

According to Proposition 2, for the fixed $\mathbf{W},\mathbf{H}$, we have

$$G(\mathbf{W},\mathbf{H})=\min_{\mathbf{u}}\hat{G}(\mathbf{W},\mathbf{H},\mathbf{u}) \quad (14)$$

and the minimum is reached at $u_l=\exp(-(\|(\mathbf{X}-\mathbf{WH})_{l,:}\|_2^2)/\sigma_1^2)$.

Then, it can be concluded that minimizing $G(\mathbf{W},\mathbf{H})$ is identical to minimizing the augmented function $\hat{G}(\mathbf{W},\mathbf{H},\mathbf{u})$ in (22)

$$\min_{\mathbf{W},\mathbf{H}}G(\mathbf{W},\mathbf{H})=\min_{\mathbf{W},\mathbf{H},\mathbf{u}}\hat{G}(\mathbf{W},\mathbf{H},\mathbf{u}). \quad (15)$$

The augmented function $\hat{G}(\mathbf{W},\mathbf{H},\mathbf{u})$ can be optimized through the following alternating minimization way.

1) Given fixed $\mathbf{W}$ and $\mathbf{H}$, minimize $\hat{G}(\mathbf{W},\mathbf{H},\mathbf{u})$ with respect to $\mathbf{u}$ by

$$u_l=\exp\left(-\frac{\|(\mathbf{X}-\mathbf{WH})_{l,:}\|_2^2}{\sigma_1^2}\right). \quad (16)$$

2) Given fixed $\mathbf{u}$, minimize $\hat{G}(\mathbf{W},\mathbf{H},\mathbf{u})$ with respect to $\mathbf{W},\mathbf{H}$. The minimization problem is equivalent to

$$\begin{aligned}&\min_{\mathbf{W},\mathbf{H}}\mathrm{Tr}((\mathbf{X}-\mathbf{WH})^T\mathbf{U}(\mathbf{X}-\mathbf{WH}))\\&\text{s.t. }\mathbf{W}\geq 0,\mathbf{H}\geq 0\end{aligned} \quad (17)$$

where $\mathrm{Tr}(\cdot)$ denotes the trace of a matrix and $\mathbf{U}$ is a diagonal matrix with diagonal element $U_{ll}=(u_l/\sigma_1^2)$. The minimization problem in (17) can be regarded as a weighted NMF problem, which can be further formulated as

$$\begin{aligned}&\min_{\mathbf{W},\mathbf{H}}\mathrm{Tr}((\hat{\mathbf{X}}-\hat{\mathbf{W}}\mathbf{H})^T(\hat{\mathbf{X}}-\hat{\mathbf{W}}\mathbf{H}))\\&\text{s.t. }\hat{\mathbf{W}}\geq 0,\ \mathbf{H}\geq 0\end{aligned} \quad (18)$$

where $\hat{\mathbf{X}}=\mathbf{U}^{1/2}\mathbf{X}$ and $\hat{\mathbf{W}}=\mathbf{U}^{1/2}\mathbf{W}$. The multiplicative update rules for (18) are given by

$$\hat{\mathbf{W}}_{lp}\leftarrow\hat{\mathbf{W}}_{lp}\frac{(\hat{\mathbf{X}}\mathbf{H}^T)_{lp}}{(\hat{\mathbf{W}}\mathbf{H}\mathbf{H}^T)_{lp}} \quad (19)$$

$$\mathbf{H}_{pn}\leftarrow\mathbf{H}_{pn}\frac{(\hat{\mathbf{W}}^T\hat{\mathbf{X}})_{pn}}{(\hat{\mathbf{W}}^T\hat{\mathbf{W}}\mathbf{H})_{pn}}. \quad (20)$$

III. Proposed Method

A. Correntropy-Based Spatial-Spectral Robust Unmixing Model

In CENMF, the noise levels of different bands are firstly explored and embedded into the unmixing process by utilizing correntropy-based metric. In hyperspectral data, there exist noisy bands with different intensities due to sensors and atmospheric effects [46], which may severely degrade the unmixing performance. In hyperspectral data, there also exist some bands not well-registered with other bands that may degrade the unmixing performance, which have been analyzed in our real hyperspectral data experiment. CENMF offers an efficient method to allocate adaptive weight for each band and suppress the influence of noisy bands. However, the diversity of the noise levels of pixels is not considered. In practical situations, there also commonly exist some pixels corrupted by a variety of degradations such as photon noise, dark noise, and fixed-pattern noise, as well as deadlines, stripes, or large DN variations caused by the sensor, which can be seen as noisy pixels [34], [36]. The existence of noisy pixels seriously reduces the performance of HU methods and may lead to a failure of finding endmembers. Since noisy bands and noisy pixels commonly exist in hyperspectral data simultaneously, we are inspired to propose a spatial-spectral robust unmixing method which achieves robustness in both the spatial dimension and spectral dimension. As shown in (7), the objective function of CENMF $G(\mathbf{W},\mathbf{H})$ applies correntropy-based metric to the residue errors by band to achieve robustness in the spectral dimension. In CENMF, a diagonal matrix $\mathbf{U}$ is introduced and updated, which contains the weight assigned for each band. To achieve robustness in the spatial dimension, we propose to apply correntropy-based metric to the residue errors by pixel and further propose a correntropy-based spatial-spectral robust unmixing model on the basis of band weights learned by CENMF. The cost function of the proposed model is defined as follows:

$$\begin{aligned}J(\mathbf{W},\mathbf{H})&=\sum_{n=1}^{N}\{-k_{\sigma_2}((\mathbf{U}^{\frac{1}{2}}\mathbf{X}-\mathbf{U}^{\frac{1}{2}}\mathbf{WH})_{:,n})\}\\&=\sum_{n=1}^{N}\left\{-\exp\left(-\frac{\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X}-\mathbf{U}^{\frac{1}{2}}\mathbf{WH})_{:,n}\|_2^2}{\sigma_2^2}\right)\right\}\end{aligned} \quad (21)$$

where $k_{\sigma_2}(\mathbf{x})=\exp(-(\|\mathbf{x}\|_2^2)/\sigma_2^2)$ and σ_2 is the kernel size. $(\mathbf{U}^{1/2}\mathbf{X}-\mathbf{U}^{1/2}\mathbf{WH})_{:,n}$ denotes the nth column of $(\mathbf{U}^{1/2}\mathbf{X}-\mathbf{U}^{1/2}\mathbf{WH})$. Note that $\mathbf{U}$ is previously decided through an individual CENMF process and fixed in the proposed model.

Through the HQ technique, the proposed model can be effectively optimized and is essential to a 2-D (spatial dimension and spectral dimension) weighted NMF, which can automatically adjust the contribution of bands and pixels and achieves robustness in both the spatial and spectral dimension.

B. Correntropy-Based Adaptive Sparsity Constraint

The phenomenon of sparsity in abundances commonly exists in HSIs, which refers to a case that the observed pixels can be represented by the mixtures of a few of endmembers [9]. Accordingly, the sparsity of abundances can be utilized as a prior in unmixing by imposing sparseness constraint such as ℓ_1 or $\ell_{1/2}$ regularizer on abundances, which helps to obtain unique solutions and lead to better results in the scenes with obvious sparsity. In [25], ℓ_1 sparse penalty is combined into CENMF to formulate ℓ_1-CENMF model to take the sparsity of abundances into consideration, of which

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the cost function is defined as follows:

$$G(\mathbf{W},\mathbf{H})=\sum_{l=1}^{L}\left\{-\exp\left(-\frac{\|(\mathbf{X}-\mathbf{WH})_{l,:}\|_2^2}{\sigma_1^2}\right)\right\}+2\lambda\sum_{n=1}^{N}\|\mathbf{H}_{:,n}\|_1 \tag{22}$$

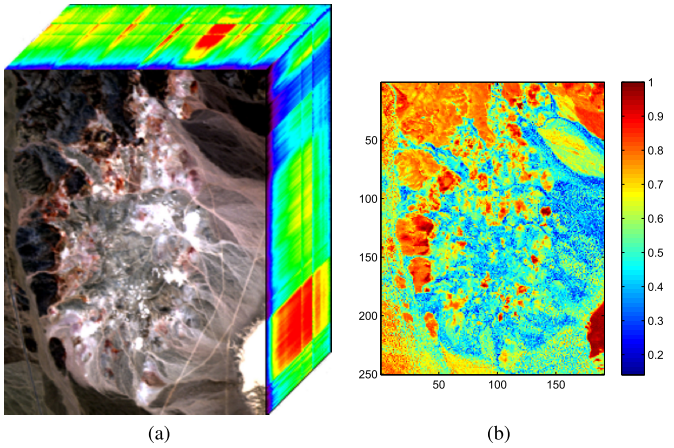
where λ is the regularization parameter which weights the contribution of the sparsity constraint. $\mathbf{H}_{:,n}$ denotes the nth column of $\mathbf{H}$. How to decide the value of the regularization parameter is a problem worthy of consideration, which greatly affects the performance of unmixing. In ℓ_1-CENMF, the parameter λ is decided by searching in an interval and evaluating the performance of algorithm in different parameter values, which is time consuming and the parameter λ is not directly decided by the sparsity level of data. In [15], Qian *et al*. propose the $\ell_{1/2}$-NMF based on the $\ell_{1/2}$ regularizer. Since the value of regularization parameter is closely related to the sparsity level of the material abundances, $\ell_{1/2}$-NMF uses an estimator for λ based on the sparseness criteria proposed in [47], which is given by

$$\lambda=\frac{1}{\sqrt{L}}\sum_{l=1}^{L}\frac{\sqrt{N}-\|\mathbf{X}_{l,:}\|_1/\|\mathbf{X}_{l,:}\|_2}{\sqrt{N}-1} \tag{23}$$

where $\mathbf{X}_{l,:}$ denotes the lth band of hyperspectral data. Apparently, the regularization parameter λ in $\ell_{1/2}$-NMF is also not directly decided by the sparsity level of abundances, but indirectly estimated by the observed hyperspectral data.

More importantly, in ℓ_1-CENMF, $\ell_{1/2}$-NMF or other unmixing methods based on sparsity constraint, the abundance vectors of pixels usually take the same value of regularization parameter, which means all the abundances of pixels are assigned with the same regularization strength. For each pixel in the hyperspectral data, the abundance vector is enforced sparsity indiscriminately. Nevertheless, it is worth noting that the sparsity levels of different regions usually vary significantly in HSIs [48]. Fig. 1 displays the well-known Cuprite data set collected by an airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the Cuprite mining site and the corresponding estimated sparseness map of this scene [48]. As shown in Fig. 1(b), there exist some regions with high sparsity levels of abundances, while there also exist some other regions with low sparsity levels since minerals are evenly mixed there. Therefore, when a sparsity constraint is imposed, applying the same regularization parameter to the entire image may not contribute to the improvement of unmixing accuracy since the sparsity levels of pixels' abundances vary. In HSIs, the regions with obvious sparsity of abundances should be emphasized with larger values of the regularization parameter, while the highly mixed regions should be assigned with smaller values.

To achieve the adaptive setting of the regularization parameter to accommodate various sparsity levels, we propose to introduce a correntropy-based metric to measure the sparsity penalty in the unmixing model. In CENMF, the correntropy-based metric is applied to the residue error of each band. Consequently, the bands with small reconstruction errors will have more weights in the optimization so that the influence of noise-corrupted bands is reduced. The automatically adaptive assignment of weight in CENMF is achieved by utilizing the property of the convex conjugated function and HQ technique. Similarly, our spatial-spectral robust model proposed in Section III-A further applies the correntropy-based metric to both the band's residue error and pixel's residue error and achieve robustness in both the spatial dimension and spectral dimension. Interestingly, in addition to being applied on the residue error part, correntropy can be also be introduced as a sparsity penalty term to achieve adaptive weighted sparsity constraint. The proposed correntropy-based adaptive sparsity constraint is defined as follows:



(a) (b)

Fig. 1. (a) AVIRIS hyperspectral data of the Cuprite mining district in NV, USA. (b) Estimated sparseness map from the obtained abundance.

$$\sum_{n=1}^{N}\exp\left(-\frac{\|\mathbf{H}_{:,n}\|_{1/2}}{\sigma_3^2}\right) \tag{24}$$

where $\|\mathbf{H}_{:,n}\|_{1/2}=\sum_{p=1}^{P}\mathbf{H}_{pn}^{1/2}$.

Through the correntropy-based sparsity penalty term in (24), each pixel is expected to be enforced with a sparsity constraint of a different weight according to its sparsity level of abundances.

C. Correntropy-Based Spatial-Spectral Robust Sparsity-Regularized (CSsRS) Unmixing Model

Combining the correntropy-based loss function in (21) with the correntropy-based sparsity constraint in (24), we further propose the following correntropy-based spatial-spectral robust sparsity-regularized (CSsRS) unmixing model:

$$\begin{aligned}J(\mathbf{W},\mathbf{H}) &= \sum_{n=1}^{N}\{-k_{\sigma_2}((\mathbf{U}^{\frac{1}{2}}\mathbf{X}-\mathbf{U}^{\frac{1}{2}}\mathbf{WH})_{:,n})\}\\ &\quad+\mu\sum_{n=1}^{N}\{-k_{\sigma_3}(\mathbf{H}_{:,n}^{\frac{1}{4}})\}\\ &=\sum_{n=1}^{N}\left\{-\exp\left(-\frac{\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X}-\mathbf{U}^{\frac{1}{2}}\mathbf{WH})_{:,n}\|_2^2}{\sigma_2^2}\right)\right\}\\ &\quad+\mu\sum_{n=1}^{N}\left\{-\exp\left(-\frac{\|\mathbf{H}_{:,n}\|_{1/2}}{\sigma_3^2}\right)\right\}\end{aligned} \tag{25}$$

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where $k_{\sigma_3}(\mathbf{x}) = \exp(-(\|\mathbf{x}\|_2^2)/\sigma_3^2)$ and μ is the regularization parameter balancing the contribution of the residue error part and the sparsity penalty term. In this article, the regularization parameters μ and λ for the proposed method and compared methods are all computed by the same strategy as (23) to make fair comparisons [15], [24].

Note that the $\ell_{1/2}$ regularizer is adopted in the proposed model since it possesses two advantages over ℓ_1 regularizer. Firstly, ℓ_1 regularizer may not have the effect of enforcing sparsity when the additivity constraint over the abundances is added since the sum-to-one constraint is a fixed ℓ_1 norm, while $\ell_{1/2}$ regularizer can still enforce sparsity with the full additivity constraint imposed. The second advantage is that the $\ell_{1/2}$ regularizer can obtain sparser solutions than the ℓ_1 regularizer does [49]. By introducing the HQ technique, the proposed model in (25) can be effectively optimized and is essentially a spatial-spectral weighted NMF with a weighted sparsity constraint. Therefore, we note the proposed method as CSsRS-NMF. Compare to CENMF, CSsRS-NMF expands its ability of robust unmixing in spatial dimension and can achieve robustness to both the band noise and pixel noise. More importantly, CSsRS-NMF incorporates a correntropy-based sparsity penalty term and is able to enforce weighted sparsity constraint with weight adaptive to the sparsity level of abundances.

D. Optimization

Since the optimization function in (25) is nonlinear, it is difficult to optimize directly. The HQ technique [50], [51] can be used to solve this correntropy-based optimization problem. HQ is a general optimization method used for convex or nonconvex minimization problems [45], which constructs an augmented cost function by introducing auxiliary variables to get the HQ reformulation of the original cost function. The optimization problem of the proposed CSsRS unmixing model in (25) can be generalized as the following problem:

$$\begin{aligned}&\min \sum_i \varphi_R(\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X}-\mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H})_{:,n}\|_2) + \sum_j \varphi_S(\|\mathbf{H}_{:,n}^{\frac{1}{4}}\|_2)\\ &\text{s.t. } \mathbf{W} \geq 0,\ \mathbf{H} \geq 0\end{aligned} \tag{26}$$

where φ_R and φ_S are the potential functions satisfy the conditions in Proposition 1. By introducing the multiplicative auxiliary variable in HQ, the problem in (26) can be reformulated as an augmented HQ optimization problem. When the auxiliary variables are given, the original problem becomes a quadratic problem, hence the name of this method. Therefore, (26) can be solved in the following alternative way:

$$v_n^k = \delta_R(\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H})_{:,n}\|_2) \tag{27}$$

$$z_n^k = \delta_S(\|\mathbf{H}_{:,n}^{\frac{1}{4}}\|_2) \tag{28}$$

$$\min\left\{\sum_{n=1} v_n\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H})_{:,n}\|_2^2 + 2\mu\sum_{n=1} z_n\|\mathbf{H}_{:,n}\|_{1/2}\right\} \tag{29}$$

where δ_R, δ_S are the minimizer functions of φ_R, φ_S respectively.

In this article, $\varphi_R = -\exp(-x^2), \varphi_S = -\exp(-x^2)$ are used as potential functions and the corresponding minimizer functions are $\delta_R = 2\exp(-x^2)$ and $\delta_S = 2\exp(-x^2)$. According to Proposition 2, substituting (12) into (26), we have the augmented objective function in an enlarged parameter space

$$\begin{aligned}\hat{J}(\mathbf{W},\mathbf{H},\mathbf{v},\mathbf{z}) = &\sum_{n=1}^{N}\left(v_n\frac{\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H})_{:,n}\|_2^2}{\sigma_2^2} + \psi_2(v_n)\right)\\ &+ 2\mu\sum_{n=1}^{N}\left(z_n\frac{\|\mathbf{H}_{:,n}\|_{1/2}}{\sigma_3^2} + \psi_3(z_n)\right)\end{aligned} \tag{30}$$

where $\mathbf{v} = [v_1, v_2, \ldots, v_N]^T, \mathbf{z} = [z_1, z_2, \ldots, z_N]^T$ are the auxiliary vectors introduced by HQ. According to Proposition 2, for the fixed $\mathbf{W}, \mathbf{H}$, the following equation holds:

$$J(\mathbf{W},\mathbf{H}) = \min_{\mathbf{v},\mathbf{z}} \hat{J}(\mathbf{W},\mathbf{H},\mathbf{v},\mathbf{z}) \tag{31}$$

and for the fixed $\mathbf{W}, \mathbf{H}$, the minimum of $\hat{J}(\mathbf{W},\mathbf{H},\mathbf{v},\mathbf{z})$ is reached at $v_n = \exp(-(\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{1/2}\mathbf{W}\mathbf{H})_{:,n}\|_2^2)/\sigma_2^2), z_n = \exp(-(\|\mathbf{H}_{:,n}\|_{1/2})/\sigma_3^2)$.

The augmented function $\hat{J}$ and the original function J share the same infimum and the same minimizer [52]. Then, it follows that

$$\min_{\mathbf{W},\mathbf{H}} J(\mathbf{W},\mathbf{H}) = \min_{\mathbf{W},\mathbf{H},\mathbf{v},\mathbf{z}} \hat{J}(\mathbf{W},\mathbf{H},\mathbf{v},\mathbf{z}). \tag{32}$$

The minimizer of $\hat{J}$ can be calculated using the following alternate minimization.

1) Given fixed $\mathbf{W}$ and $\mathbf{H}$, minimize $\hat{J}(\mathbf{W},\mathbf{H},\mathbf{v},\mathbf{z})$ with respect to $\mathbf{v}, \mathbf{z}$,

$$v_n = \exp\left(-\frac{\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H})_{:,n}\|_2^2}{\sigma_2^2}\right) \tag{33}$$

$$z_n = \exp\left(-\frac{\|\mathbf{H}_{:,n}\|_{1/2}}{\sigma_3^2}\right). \tag{34}$$

2) Given fixed $\mathbf{v}, \mathbf{z}$, minimize $\hat{J}(\mathbf{W},\mathbf{H},\mathbf{v},\mathbf{z})$ with respect to $\mathbf{W}, \mathbf{H}$

$$\begin{aligned}&\min_{\mathbf{W},\mathbf{H}} \sum_{n=1}^{N} v_n\frac{\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H})_{:,n}\|_2^2}{\sigma_2^2} + 2\mu\sum_{n=1}^{N} z_n\frac{\|\mathbf{H}_{:,n}\|_{1/2}}{\sigma_3^2}\\ &\text{s.t. } \mathbf{W} \geq 0, \quad \mathbf{H} \geq 0.\end{aligned} \tag{35}$$

Apparently, the optimization problem in (35) can be seen as weighted NMF problem with weighted $\ell_{1/2}$ regularizer and can be further formulated as

$$\begin{aligned}&\min_{\mathbf{W},\mathbf{H}} \operatorname{Tr}(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H})\mathbf{V}(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H})^T\\ &\qquad + 2\mu\sum_{n=1}^{N}\frac{z_n}{\sigma_3^2}\|\mathbf{H}_{:,n}\|_{1/2}\\ &\text{s.t. } \mathbf{W} \geq 0, \quad \mathbf{H} \geq 0\end{aligned} \tag{36}$$

where $\mathbf{V}$ is a diagonal matrix with diagonal element $\mathbf{V}_{nn} = (v_n/\sigma_2^2)$.

Let $\Theta \in \mathbb{R}^{L\times P}$ and $\Xi \in \mathbb{R}^{P\times N}$ be the corresponding Lagrange multipliers for the constraint $\mathbf{W} \geq 0$ and $\mathbf{H} \geq 0$ respectively. Consider the Lagrange function as

$$\begin{aligned}\mathcal{L} = &\operatorname{Tr}((\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H})\mathbf{V}(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{W}\mathbf{H}))^T\\ &+ 2\mu\sum_{n=1}^{N}\frac{z_n}{\sigma_3^2}\|\mathbf{H}_{:,n}\|_{1/2} + \operatorname{Tr}(\Theta\mathbf{W}) + \operatorname{Tr}(\Xi\mathbf{H}).\end{aligned} \tag{37}$$

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Take the partial derivatives of $\mathcal{L}$ with respect to $\mathbf{W}_{lp}$ and $\mathbf{H}_{pn}$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = -2\mathbf{UXVH}^T + 2\mathbf{UWHVH}^T + \Theta \tag{38}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{H}} = -2\mathbf{W}^T\mathbf{UXV} + 2\mathbf{W}^T\mathbf{UWHV} + 2\mu\mathbf{H}^{-\frac{1}{2}}\mathbf{Z} + \Xi \tag{39}$$

where $\mathbf{Z}$ is a diagonal matrix with diagonal element $\mathbf{Z}_{nn} = (z_n/\sigma_3^2)$.

According to the Karush–Kuhn–Tucker conditions $\Theta_{lp}\mathbf{W}_{lp} = 0$ and $\Xi_{pn}\mathbf{H}_{pn} = 0$, we have

$$(-\mathbf{UXVH}^T + \mathbf{UWHVH}^T).*\mathbf{W} = 0 \tag{40}$$

$$(-\mathbf{W}^T\mathbf{UXV} + \mathbf{W}^T\mathbf{UWHV} + \mu\mathbf{H}^{-\frac{1}{2}}\mathbf{Z}).*\mathbf{H} = 0. \tag{41}$$

Then, the multiplicative updating rules can be derived as follows:

$$\mathbf{W}_{lp} \leftarrow \mathbf{W}_{lp}\frac{(\mathbf{XVH}^T)_{lp}}{(\mathbf{WHVH}^T)_{lp}} \tag{42}$$

$$\mathbf{H}_{pn} \leftarrow \mathbf{H}_{pn}\frac{(\mathbf{W}^T\mathbf{UXV})_{pn}}{(\mathbf{W}^T\mathbf{UWHV} + \mu\mathbf{H}^{-\frac{1}{2}}\mathbf{Z})_{pn}}. \tag{43}$$

Note that the diagonal matrix $\mathbf{U}$ containing weights of bands only works on the updating of $\mathbf{H}$, while the diagonal matrix $\mathbf{V}$ containing weights of pixels works on both the updating of $\mathbf{W}$ and $\mathbf{H}$ due to the existence of regularization term of $\mathbf{H}$. Actually, the updating rule of CENMF in (19) can also be simplified since the weights on bands would not interfere with the updating of $\mathbf{W}$.

As the above optimization procedure shows, the nonlinear optimization problem in (25) can be solved effectively through the HQ technique. By introducing auxiliary variables, the constructed augmented cost function shares the same infimum and minimizer with the original cost function and the optimization of the two is equivalent. Therefore, the proposed algorithm consists of computing the auxiliary variables and solving the problem of weighted NMF with a weighted sparsity constraint. The detailed optimization procedure of our proposed algorithm is described in Algorithm 1.

E. Implementation Issues

The first issue we are concerned about is the initialization method. $\mathbf{W}$ and $\mathbf{H}$ are firstly initialized by SGA-FCLS. And $\mathbf{U}$ is computed by applying CENMF. To eliminate the interference of noisy bands, we adopt an effective initialization strategy to generate initial values for $\mathbf{H}$ by utilizing the weight information contained in $\mathbf{U}$. Specifically, $\mathbf{W}$ is reinitialized by applying SGA on $\mathbf{X}$, and $\mathbf{W}, \mathbf{U}$ are used to reinitialize $\mathbf{H}$ by $\mathbf{H} = \mathrm{FCLS}(\mathbf{U}^{1/2}\mathbf{X}, \mathbf{U}^{1/2}\mathbf{W})$.

Another issue we are concerned about is the parameter setting. Similar to the kernel size σ_1 in (8), the kernel size σ_2, σ_3 are computed by

$$\sigma_2 = \frac{\beta}{2N}\sum_{n=1}^{N}\|(\mathbf{U}^{\frac{1}{2}}\mathbf{X} - \mathbf{U}^{\frac{1}{2}}\mathbf{WH})_{:,n}\|_2^2 \tag{44}$$

$$\sigma_3 = \frac{\beta}{2N}\sum_{n=1}^{N}\|\mathbf{H}_{:,n}\|_{\frac{1}{2}} \tag{45}$$

Algorithm 1 CSsRS-NMF Algorithm

Input: The observed mixture data $\mathbf{X}$; the number of endmembers P.

Output: Endmembers matrix $\mathbf{W}$ and abundance matrix $\mathbf{H}$.

```
Initialize endmember matrix W and abundance matrix H by SGA-FCLS.
Conduct CENMF by (16, 19, 20) and obtain U.
Reinitialize W and H.
repeat
    Compute σ2 according to (44).
    Compute σ3 according to (45).
    Update V by (33).
    Update Z by (34).
    Update W by (42).
    Update H by (43).
until convergence
return W and H
```

where β is set to 1 throughout the article. In this article, the regularization parameters μ for the proposed CSsRS-NMF and λ for the compared methods are all decided by the same strategy as (23) to make fair comparisons.

IV. Experiments

A. Experiments on Synthetic Data

In this section, the proposed CSsRS-NMF is tested on synthetic data to evaluate its performance. Five related methods, including SGA-FCLS, MVCNMF [18], $\ell_{2,1}$-NMF [32], ℓ_1-CENMF [25], and $\ell_{1/2}$-RNMF [24] are compared with the proposed method in the experiments. The abundances of synthetic data are generated following the method in [15]. A threshold θ called purity level is used to control the sparsity level of synthetic data. The pixels with abundance larger than θ will be replaced by the mixtures of all endmembers with equal abundance. A smaller value of θ produces synthetic data with lower sparsity level, which means the synthetic data are mixed in a higher degree. The spectral signatures used as endmembers are randomly selected from the United States Geological Survey (USGS) digital spectral library [53], which are composed of 224 spectral bands ranging from 0.38 to 2.5 μm with spectral resolution of 10 nm. Six sample spectral signatures are displayed in Fig. 2. In the real-world applications, both noisy bands and noisy pixels commonly exist in the HSIs and the noise levels of different bands or different pixels vary. And there also exist signal-dependent (SD) noise which varies with variance dependent on the signal level. To get better simulation of hyperspectral data consistent with the real-world situations, the synthetic data are generated in two ways to validate the performance of algorithms.

In the first way, we aim to investigate the performance of algorithm on noise with the forms of band noise and pixel noise. Both band noise and pixel noise are added to the synthetic data, and the noise intensity for different band or pixel is different, which means the SNR value for each band

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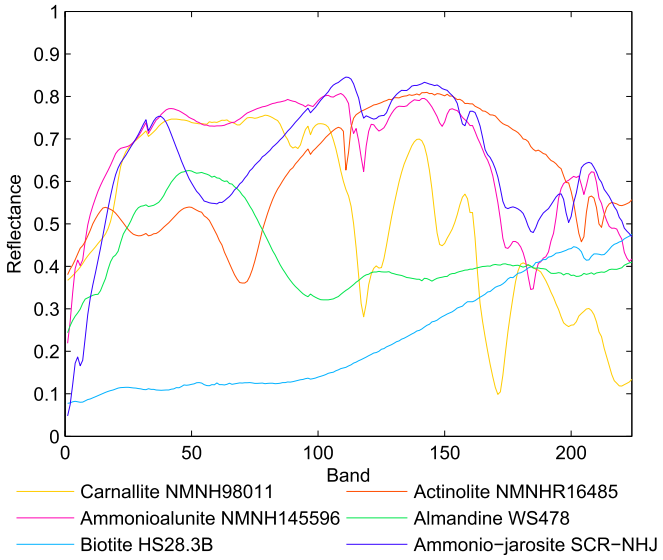


Fig. 2. Sample spectral signatures from the USGS library.

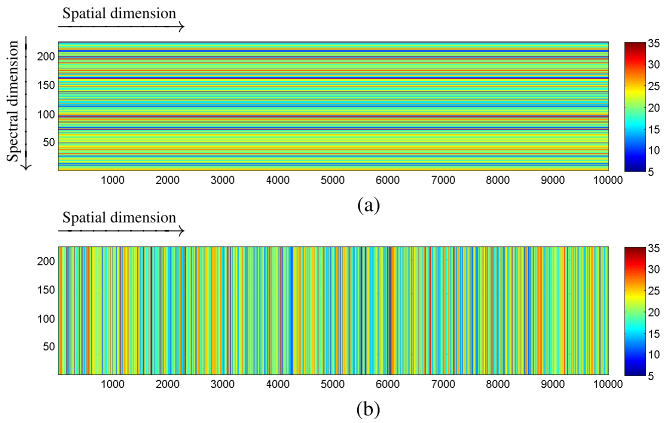


Fig. 3. SNR values for pixel noise and band noise. (a) SNR value for each band. (b) SNR value for each pixel.

or pixel is different. The SNR is defined as follows:

$$\mathrm{SNR} = 10\log_{10}\frac{E[\mathbf{x}^T\mathbf{x}]}{E[\mathbf{n}^T\mathbf{n}]} \tag{46}$$

where $\mathbf{x}$ and $\mathbf{n}$ denote the observation and noise of a pixel or a band, respectively, and $E[\cdot]$ denotes the expectation operator. Specifically, band noise and pixel noise are generated as follows.

1) *Band noise:* We add zero-mean Gaussian noise to each band of the HSI. For different bands, the levels of noise intensities are different. The SNR values of all the bands are sampled from a Gaussian distribution $\mathrm{SNR}_{\mathrm{band}} \sim \mathcal{N}(\overline{\mathrm{SNR}}_{\mathrm{band}}, \sigma^2)$. σ is fixed as 5 in this article. When $\overline{\mathrm{SNR}}_{\mathrm{band}}$ is set to 20, Fig. 3(a) presents the distribution of SNR value for each band, which varies in the spectral dimension.
2) *Pixel noise:* Similarly, we also add zero-mean Gaussian noise to each pixel of the HSI. For different pixels, the levels of noise intensities are different. The SNR values of all the pixels are sampled from $\mathrm{SNR}_{\mathrm{pixel}} \sim \mathcal{N}(\overline{\mathrm{SNR}}_{\mathrm{pixel}}, \sigma^2)$. When $\overline{\mathrm{SNR}}_{\mathrm{pixel}}$ is set to 20, Fig. 3(b) presents the distribution of SNR value for each pixel, which varies in the spatial dimension.

In the second way of generating synthetic data, a noisy HSI is described by the following model [54]–[56]:

$$x(i,j,l) = g(i,j,l) + n_{si}(i,j,l) + \sqrt{g(i,j,l)}\,n_{sd}(i,j,l) \tag{47}$$

where (i, j) is the spatial location of a pixel and l denotes the band index. $x(i, j, l)$ denotes the observed noisy signal and $g(i, j, l)$ denotes the noise-free signal. The additive term $\sqrt{g(i, j, l)}n_{sd}(i, j, l)$ is the SD component, while $n_{si}(i, j, l)$ is the signal-independent (SI) component. $n_{si}(i, j, l)$ and $n_{sd}(i, j, l)$ are zero-mean Gaussian-distributed random values with variance σ_{si}^2 and σ_{sd}^2. To set the values of σ_{si}^2 and σ_{sd}^2, the following quantities are defined [54]:

$$\phi = \left(\frac{1}{\mathrm{SNR}_{si}} + \frac{1}{\mathrm{SNR}_{sd}}\right)^{-1} \tag{48}$$

$$\mathrm{SNR}_{si} = \frac{\overline{P_g}}{\sigma_{si}^2} \tag{49}$$

$$\mathrm{SNR}_{sd} = \frac{\overline{P_g}}{\sigma_{sd}^2 \cdot \overline{g}} \tag{50}$$

where $\overline{P_g} = 1/ijl\sum_{i,j,l} g(i, j, l)^2$ and $\overline{g} = 1/ijl\sum_{i,j,l} g(i, j, l)$. A coefficient α is defined by $\mathrm{SNR}_{si} = \alpha\mathrm{SNR}_{sd}$ to adjust the contribution of SD and SI component in the noise part.

Two criteria, spectral angle distance (SAD) and root-mean-square error (RMSE), are adopted to access the accuracy of estimated endmembers and abundances, which are defined as follows:

$$\mathrm{SAD}_p = \cos^{-1}\left(\frac{\mathbf{w}_p^T\hat{\mathbf{w}}_p}{\|\mathbf{w}_p\|\|\hat{\mathbf{w}}_p\|}\right) \tag{51}$$

$$\mathrm{rmsE}_p = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(h_{pn} - \hat{h}_{pn})^2} \tag{52}$$

where $\mathbf{w}_p$ and $\hat{\mathbf{w}}_p$ are the reference endmember signatures and the estimated signatures, respectively. And h_{pn} and $\hat{h}_{pn}$ are the reference abundances and their estimates. The mean values of SAD_p and rmsE_p are denoted as $\overline{\mathrm{SAD}}$ and $\overline{\mathrm{RMSE}}$, respectively.

Five experiments are conducted to validate the effectiveness of the proposed algorithm. In Experiments 1–4, the synthetic data are generated through the first way of generating synthetic data. In Experiment 5, the synthetic data are generated through the second way.

1) Experiment 1 (Robustness analysis to different noise forms): In this experiment, we investigate the performance of algorithms on synthetic data with different forms of noise added. The number of endmembers is fixed as 5. The size of data used here and in the following experiments is 100×100, except in Experiment 4. The purity level θ is set to 0.8 in all the experiments except in Experiment 2. Three kinds of

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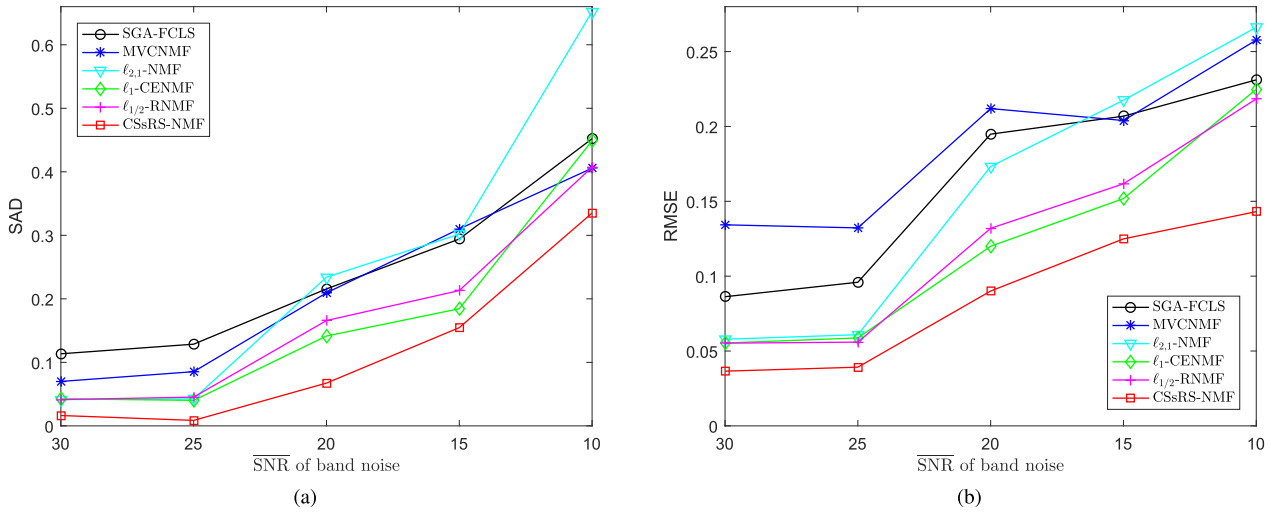


Fig. 4. Performance comparison of the algorithms with different levels of band noise. (a) $\overline{\text{SAD}}$. (b) $\overline{\text{RMSE}}$.

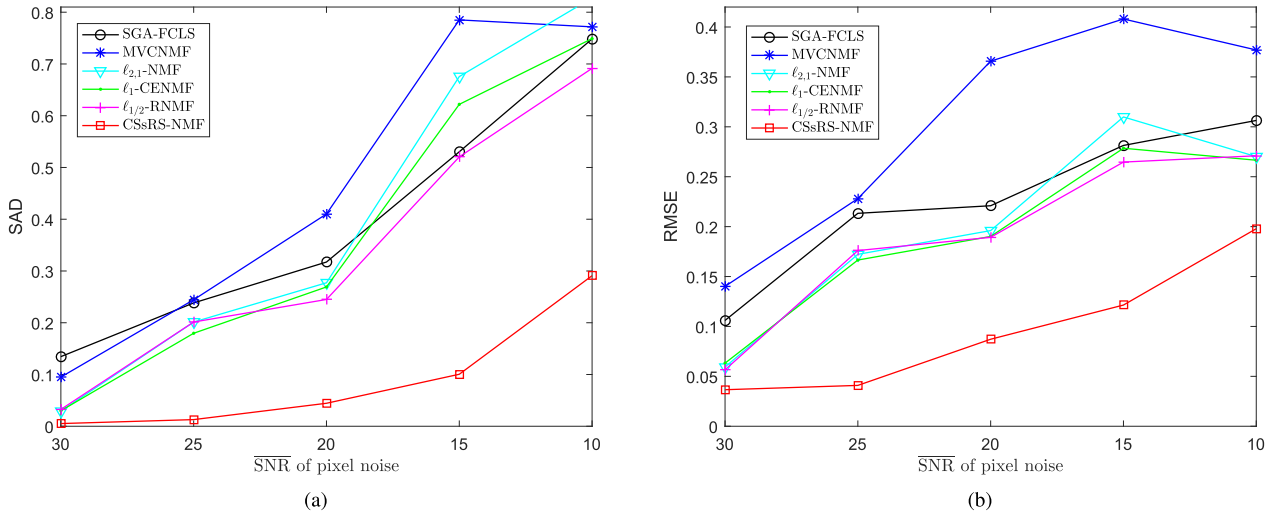


Fig. 5. Performance comparison of the algorithms with different levels of pixel noise. (a) $\overline{\text{SAD}}$. (b) $\overline{\text{RMSE}}$.

situations are considered: only band noise added, only pixel noise added, both band noise and pixel noise added.

For the first situation, each band is added with Gaussian noise of different intensity. The SNR values of all the bands are sampled from a Gaussian distribution $\text{SNR}_{\text{band}} \sim \mathcal{N}(\overline{\text{SNR}}_{\text{band}}, \delta^2)$. $\overline{\text{SNR}}_{\text{band}}$ is set to 10, 15, 20, 25, and 30 dB, respectively, to test the algorithms' performance under different levels of band noise. Fig. 4 shows the $\overline{\text{SAD}}$ and $\overline{\text{RMSE}}$ values of algorithms in the cases of different noise levels. The results demonstrate that the proposed CSsRS-NMF performs the best and has good robustness to band noise of various intensities. ℓ_1-CENMF and $\ell_{1/2}$-RNMF also show robustness to band noise as expected, especially when the values of SNR is relatively low. ℓ_1-CENMF performs slightly better than $\ell_{1/2}$-RNMF except for the case of $\overline{\text{SNR}} = 10$.

For the second situation with pixel noise added, each pixel is added with the Gaussian noise of different intensities. The SNR values of all the pixels are sampled from a Gaussian distribution $\text{SNR}_{\text{pixel}} \sim \mathcal{N}(\overline{\text{SNR}}_{\text{pixel}}, \delta^2)$. $\overline{\text{SNR}}_{\text{pixel}}$ is set to 10, 15, 20, 25, and 30 dB, respectively to test the algorithms' performance under different levels of pixel noise. The $\overline{\text{SAD}}$ and $\overline{\text{RMSE}}$ results are presented in Fig. 5. We can see that the proposed CSsRS-NMF also achieves the best results for either $\overline{\text{SAD}}$ or $\overline{\text{RMSE}}$ when the intensity of pixel noise varies, which indicates that CSsRS-NMF also has good robustness to pixel noise. $\ell_{1/2}$-RNMF also shows some robustness to pixel noise as expected. As designed, $\ell_{2,1}$-NMF should possess robustness to noise and outliers in the sample dimension (by pixel). However, the results of $\ell_{2,1}$-NMF show no obvious superiority to the other methods' results. This phenomenon is mainly caused by its unreasonable weight assignment. In $\ell_{2,1}$-NMF, the weights for pixels are directly proportional to the inverses of pixels' residues and the objective function may be dominated by the data points with small residues. The unreasonable strategy of weight assignment in $\ell_{2,1}$-NMF omits the important information of other data points and may cause that the algorithms perform unsatisfactorily when pixel noise is added.

For the third situation, band noise and pixel noise are added simultaneously to test the performance of algorithms. Each band is added with the Gaussian noise of different intensities. The SNR values of all the bands are sampled from a Gaussian distribution $\text{SNR}_{\text{band}} \sim \mathcal{N}(\overline{\text{SNR}}_{\text{band}}, \delta^2)$. And then each pixel is added with Gaussian noise of different intensity. The SNR

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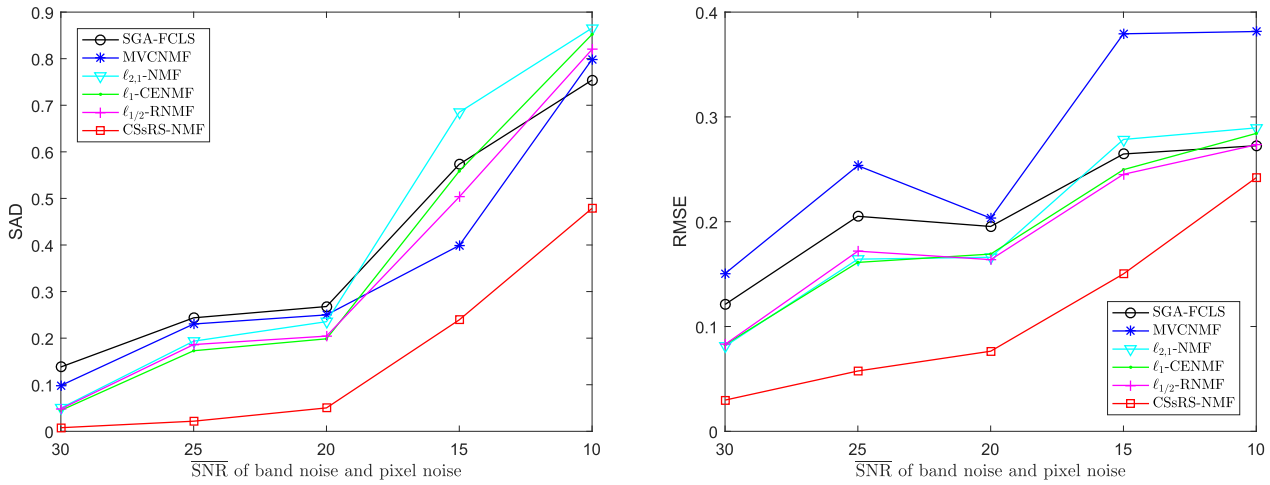


Fig. 6. Performance comparison of the algorithms with different levels of band noise and pixel noise. (a) $\overline{\text{SAD}}$. (b) $\overline{\text{RMSE}}$.

values of all the pixels are sampled from a Gaussian distribution $\text{SNR}_{\text{pixel}} \sim \mathcal{N}(\overline{\text{SNR}}_{\text{pixel}}, \delta^2)$. Both $\overline{\text{SNR}}_{\text{band}}$, $\overline{\text{SNR}}_{\text{pixel}}$ are set to 10, 15, 20, 25, and 30 dB, respectively, to test the algorithms' performance under different levels of band noise and pixel noise. The $\overline{\text{SAD}}$ and $\overline{\text{RMSE}}$ results are presented in Fig. 6. It can be observed that the proposed CSsRS-NMF outperforms other methods significantly, which is mainly due to the CSsRS-NMF's robustness in both spatial dimension and spectral dimension as well as the adaptive weighted sparsity constraint. CSsRS-NMF achieves robustness by allocating adaptive weight to bands and pixels with different noise levels. To further demonstrate the behavior of the proposed method, Fig. 7(a) presents the weight assignment for bands in the proposed method, along with the actual SNR values for the band noise added on the synthetic data. As shown in the figure, the weights assigned for bands coincide with the distribution of band noise levels well. The bands with high noise intensities are assigned with relatively low weights, while the bands with high SNR values are assigned with high weights to emphasize their contribution. Fig. 7(b) presents the weight assignment for pixels in the proposed method as well as the actual SNR values for pixel noise added on the synthetic data. It also can be seen that the weights assigned for pixels are in good agreement with the distribution of pixel noise levels. The above observations further demonstrate that the proposed CSsRS-NMF can precisely allocate weights for bands and pixels according to their noise levels and achieve robustness to band noise and pixel noise simultaneously.

In the following, Experiments 2–4, both band noise and pixel noise are added. $\overline{\text{SNR}}_{\text{band}}$ and $\overline{\text{SNR}}_{\text{pixel}}$ are set to 20.

2) Experiment 2 *(Performance comparison when the sparsity level varies):* In this experiment, we test the performance of algorithms on data with different sparsity levels. The purity level θ is set to $0.6, 0.7, 0.8, 0.9, 1$, respectively to reject pixels with high purity and obtain data with different sparsity levels. The number of endmembers is fixed as 5. Fig. 8 shows the $\overline{\text{SAD}}$ and $\overline{\text{RMSE}}$ performance of algorithms under different sparsity levels. When applied to data with various sparsity levels, $\ell_{1/2}$-RNMF achieves slightly better performance than other compared algorithms with including $\ell_{1/2}$ sparsity constraint. The proposed CSsRS-NMF achieves the best results of $\overline{\text{SAD}}$ and $\overline{\text{RMSE}}$ for all the cases, which validates that CSsRS-NMF can promote sparsity adaptively and improve unmixing performance through the proposed correntropy-based adaptive sparsity constraint.

3) Experiment 3 *(Performance comparison when the number of endmembers varies):* In this experiment, synthetic data made up of a different number of endmembers are used to examine the performance of algorithms. The number of endmembers varies from 3 to 8. The results are as shown in Fig. 9. It can be observed that CSsRS-NMF attains great advantage over other methods in both $\overline{\text{SAD}}$ and $\overline{\text{RMSE}}$.

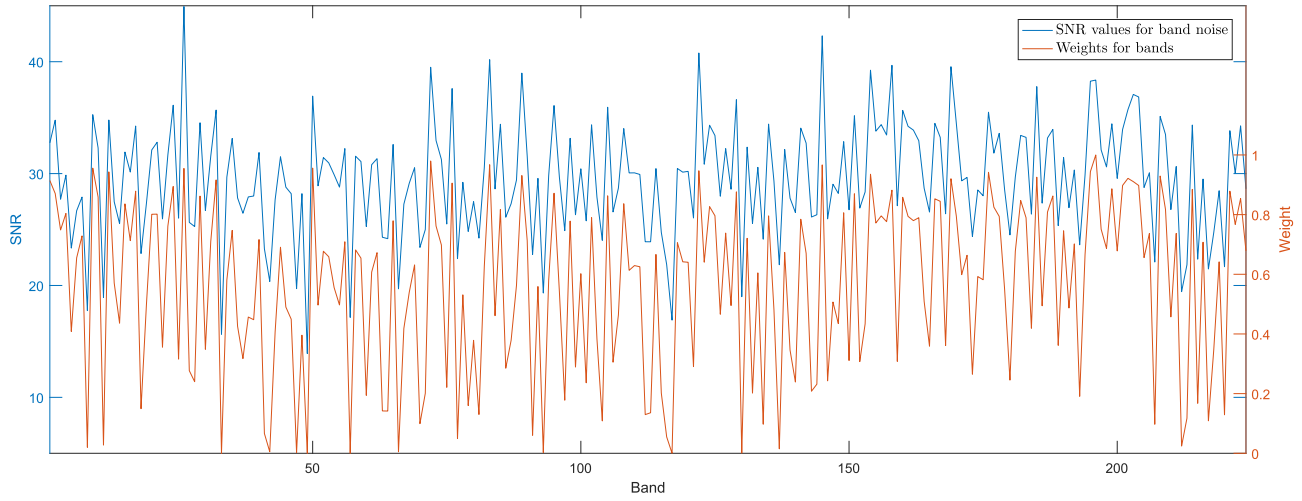
4) Experiment 4 *(Generation to the size of data):* We also vary the size of synthetic data to examine the algorithms' performance. The size of synthetic data is set to 36×36, 49×49, 64×64, 81×81, 100×100, respectively. As Fig. 10 shows, with the size of data increases, the performance of algorithms has the trend of degrading. CSsRS-NMF also achieves the lowest $\overline{\text{SAD}}$ and $\overline{\text{RMSE}}$ values, which illustrates that the proposed CSsRS-NMF can provide reliable and accurate unmixing results when applied to data of different sizes.

5) Experiment 5 *(Performance comparison when the coefficient α varies:* In this experiment, synthetic data are simulated with SD noise considered. ϕ is set to 20. The coefficient α is set to 1/4, 1, 4 respectively to simulate the following three different cases. 1) Case 1: $\alpha = 1/4$, the SI noise component is dominant; 2) Case 2: $\alpha = 1$, there is no dominant noise source. 3) Case 3: $\alpha = 4$, the SD noise component is dominant. As shown in Fig. 11, CSsRS-NMF also achieves the best estimation of endmembers and abundances in all the three cases, which demonstrates the proposed CSsRS-NMF has good robustness to both the SI noise and SD noise.

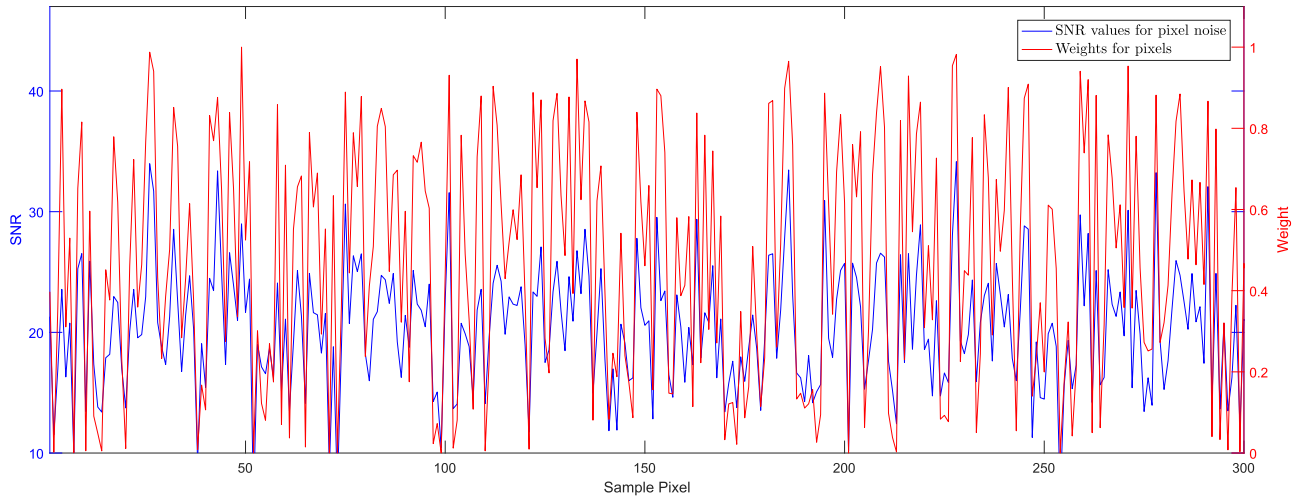
B. Experiments on Real Data

In this section, we test the unmixing performance of the proposed method on real hyperspectral data. The unmixing

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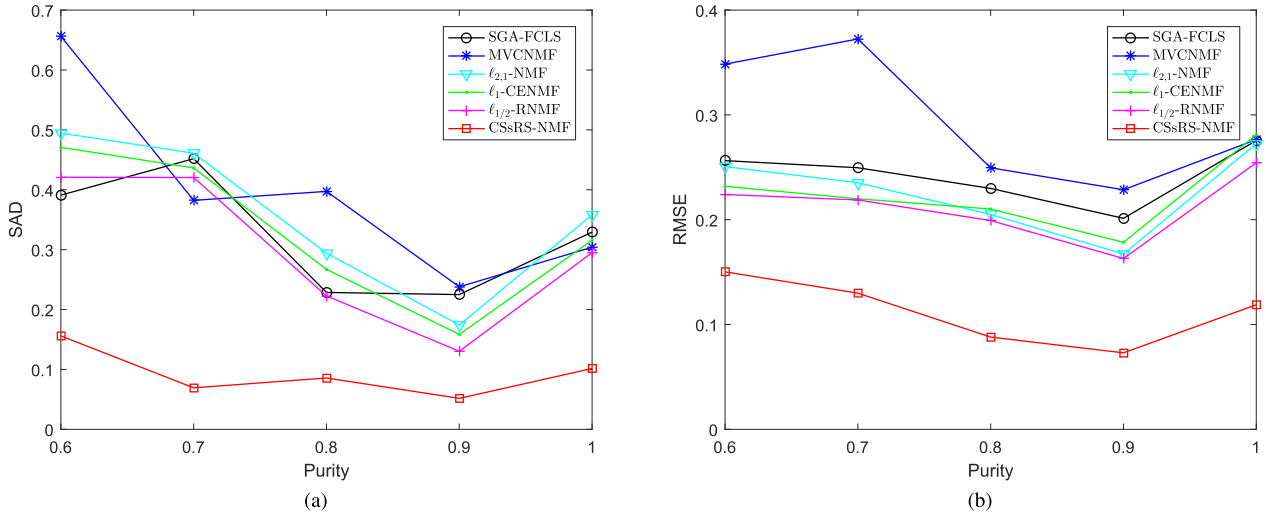


(a)



(b)

Fig. 7. Distribution of SNR values and the weight allocation in spectral dimension and spatial dimension. (a) Spectral dimension. (b) Spatial dimension.



(a) (b)

Fig. 8. Performance comparison of the algorithms with different mixing degrees. (a) $\overline{\text{SAD}}$. (b) $\overline{\text{RMSE}}$.

results of MVCNMF [18], $\ell_{2,1}$-NMF [32], ℓ_1-CENMF [25] and $l_{1/2}$-RNMF [24] are also presented to make comparisons. The endmembers found by N-FINDR [2] are also used to make comparisons. Since the ground truth of abundances is usually unavailable in the real hyperspectral experiments, we use the benchmark maps provided in [57] for reference.

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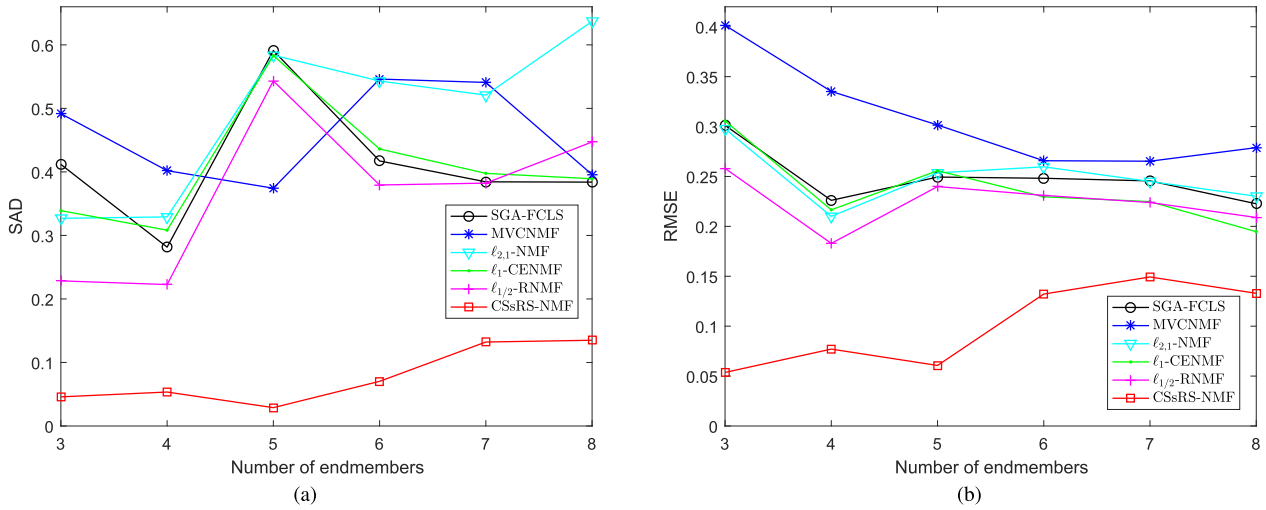


Fig. 9. Performance comparison of the algorithms with different numbers of endmembers. (a) $\overline{\text{SAD}}$. (b) $\overline{\text{RMSE}}$.

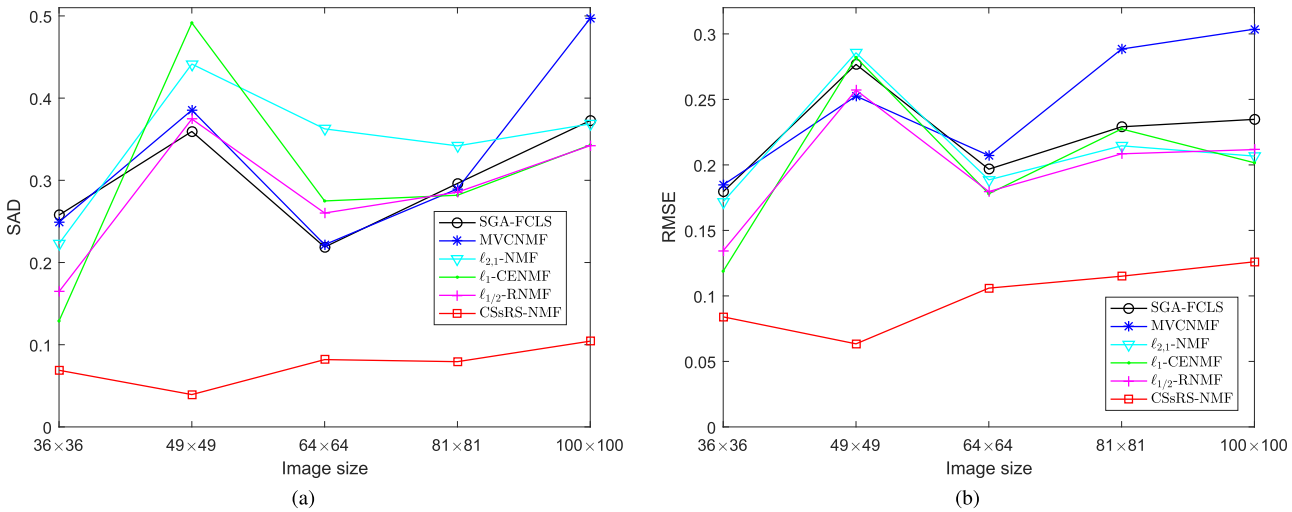


Fig. 10. Performance comparison of the algorithms with different image sizes. (a) $\overline{\text{SAD}}$. (b) $\overline{\text{RMSE}}$.

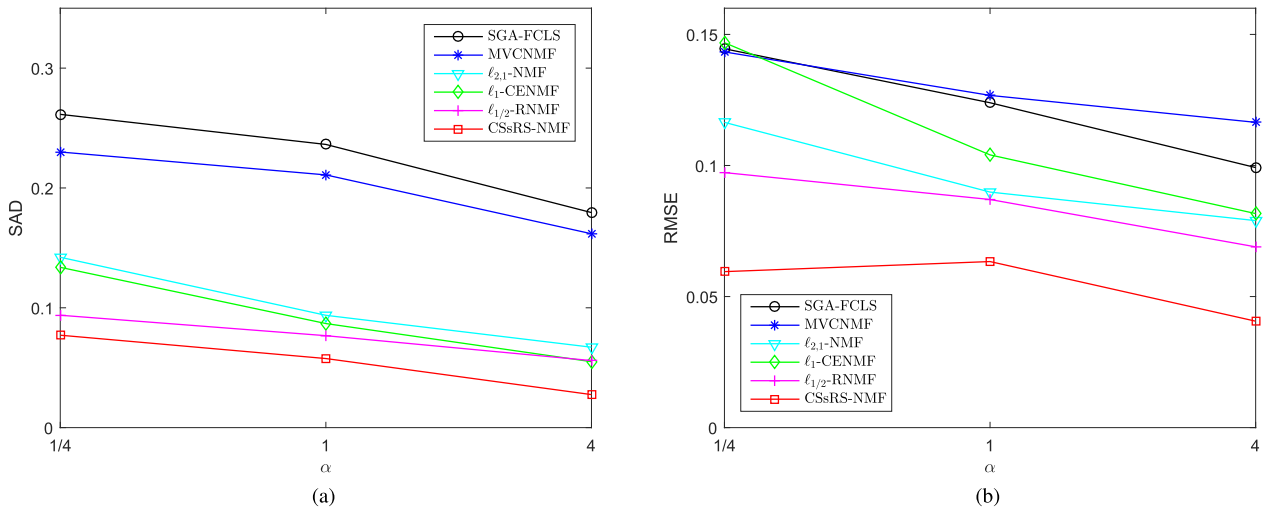
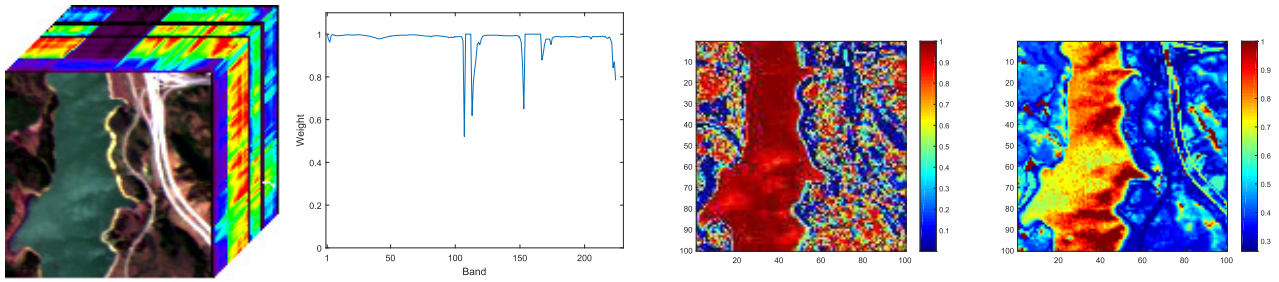


Fig. 11. Performance comparison of the algorithms with respect to α. (**a**) $\overline{\text{SAD}}$. (**b**) $\overline{\text{RMSE}}$.

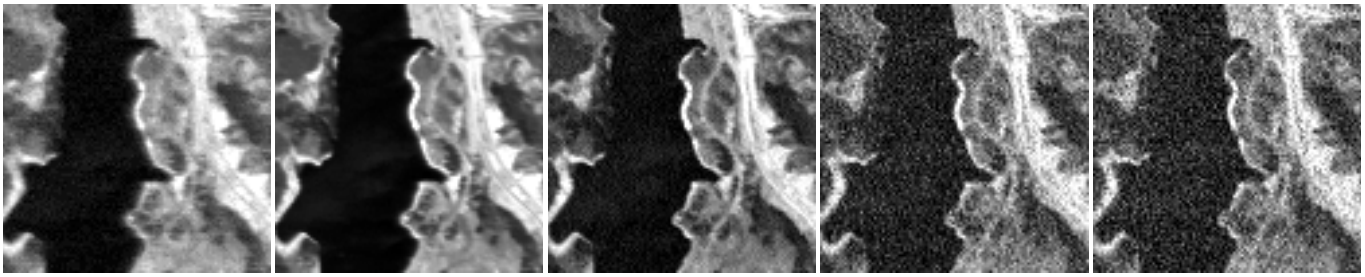
The first used hyperspectral data in the experiment are the Jasper data set popularly used in [25], [58], and [59], which is collected by an AVIRIS sensor. A 100×100 pixel subset image as shown in Fig. 12(a) is used in this experiment. The Jasper data set consists of 224 spectral bands with the wavelength ranging from 0.38 to 2.5 μm. Among these bands, there exist some blank bands, including the bands 1, 108-112, and 154–166. All 224 bands are used for the experiment to test the

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(a) (b) (c) (d)

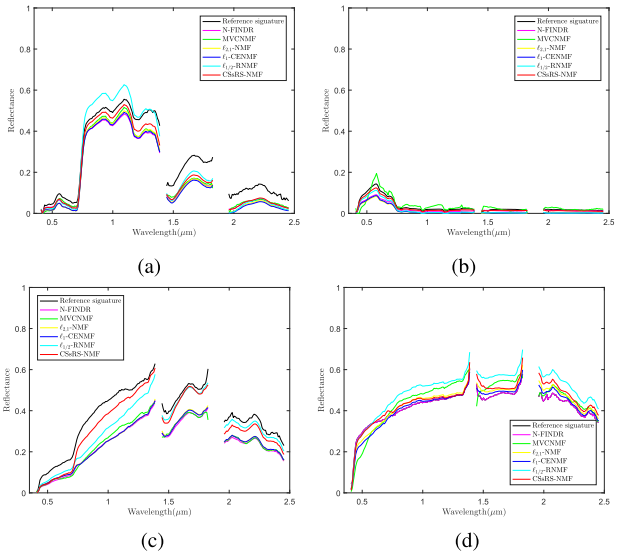
Fig. 12. Weight assignment of band, pixel, and sparsity constraint when applying CSsRS-NMF on the Jasper data set. (a) Jasper data set. (b) Weight for each band. (c) Weight for each pixel. (d) Weight for sparsity constraint of each pixel's abundances.



(a) (b) (c) (d) (e)

Fig. 13. Images of some bands in the Jasper data set. (a) Band 113 (Weight = 0.6194). (b) Band 117 (Weight = 0.9516). (c) Band 220 (Weight = 0.9768). (d) Band 222 (Weight = 0.8428). (e) Band 224 (Weight = 0.7834).

performance of algorithms. Fig. 12(b)–(d) presents the weight u_l, v_n, z_n for each band, pixel, and sparsity constraint of pixel's abundance assigned by CSsRS-NMF in the last iteration, respectively. As shown in Fig. 12(b), the weights for the blank bands are 1 since the residues $\|(\mathbf{X}-\mathbf{WH})_{l,:}\|_2$ are 0. Although these blank bands own the maximum weight, they do not contribute to the cost function since the corresponding values of $\mathbf{X}_{l,:}$, $\mathbf{W}_{l,:}$ are 0 and $\|(\mathbf{X}-\mathbf{WH})_{l,:}\|_2 = 0$. It can be observed that bands 107, 113–116, 152–153, 167–169, 222–224 are assigned with relatively low weights. To further investigate the behavior of the proposed algorithm, we present some band images of the Jasper data set in Fig. 13. As expected, the noisy bands with low SNR values are assigned with low weights accordingly. From Fig. 13(c)–(e), it can be observed that the SNR values of band images ($\mathrm{SNR}_{220}>\mathrm{SNR}_{222}>\mathrm{SNR}_{224}$) are in good agreement with the assigned weights ($0.9768 > 0.8428 > 0.7834$). Meanwhile, some bands with relatively high SNR values are assigned with low weights, including bands 107, 113–116, 152–153. Through visual inspection and comparison of Fig. 13(a) and (b), it can be found that there exists at least 1 pixel of registration error between band 113 and band 117 in the horizontal direction. This phenomenon explains why band 113 and other high SNR bands are assigned with low weights. Due to the unsatisfactory registration, these bands have high reconstruction errors and may degrade the unmixing performance. In the proposed CSsRS-NMF, these not well-registered bands are assigned with low weights, which further demonstrates the robustness of CSsRS-NMF in the spectral dimension. Fig. 12(c) presents the weights assigned for pixels by CSsRS-NMF. The regions with good reconstruction are assigned with high weights, such as the water area, while the regions with brighter pixels such as the road area are assigned with lower weights. These pixels with high intensities of signal are supposed to carry noise of higher intensities compared to other darker pixels. With the increase of image intensity, the noise of pixels' signal become SD since the noise are mainly dominated by SD noise such as photon noise and fixed-pattern noise [36]. Photon noise is related to the inherent natural variation of the photon flux arriving at the sensor. Fixed-pattern noise arises from the differences in the



(a) (b) (c) (d)

Fig. 14. Comparison of endmember signatures estimated by different methods on the Jasper data set. (a) Tree. (b) Water. (c) Soil. (c) Road.

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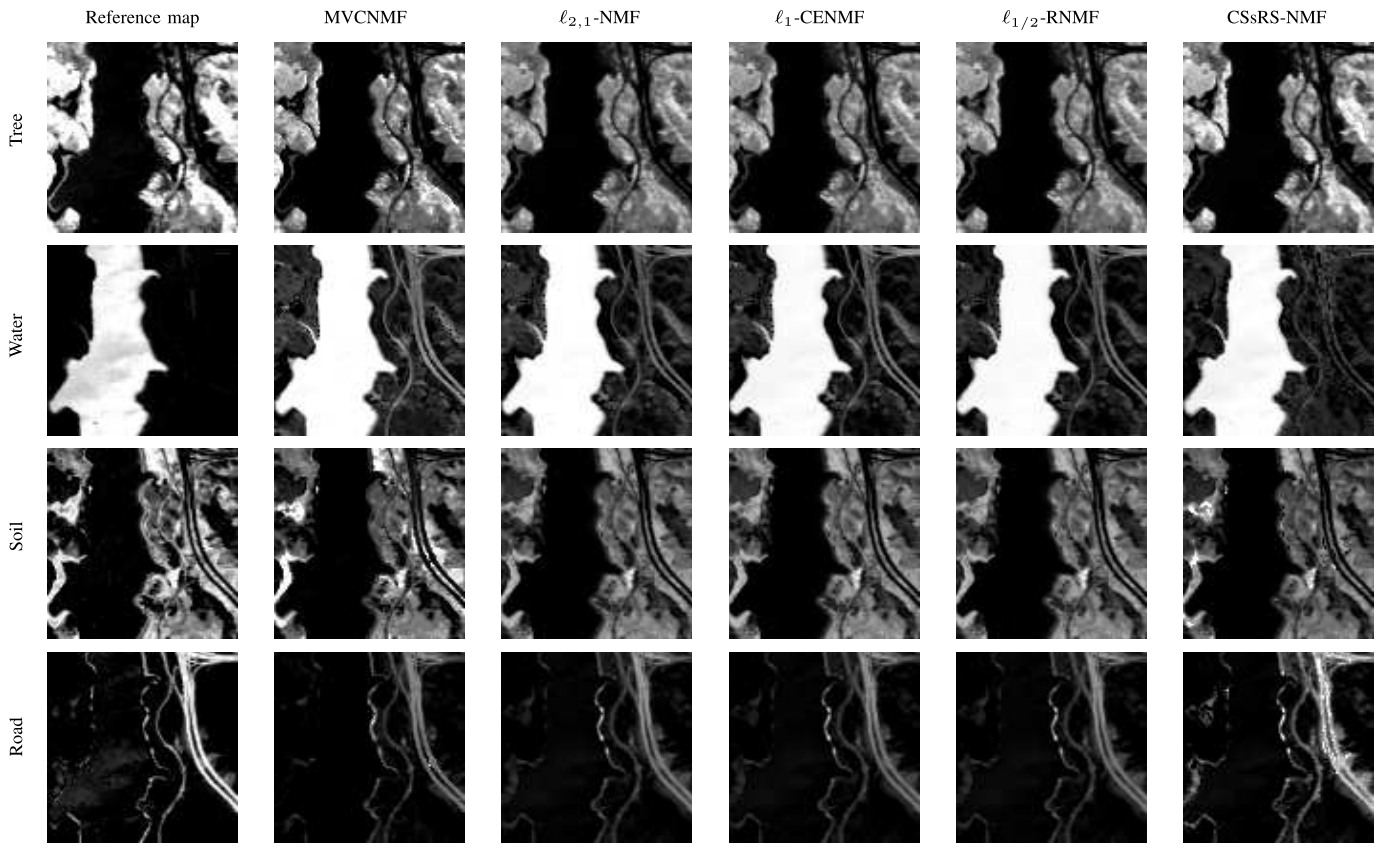


Fig. 15. Abundance maps of the Jasper data by MVCNMF, $\ell_{2,1}$-NMF, ℓ_1-CENMF, $\ell_{1/2}$-RNMF, CSsRS-NMF, along with the reference maps. Each row shows the corresponding abundance maps of the same endmember.

individual responsitivities of pixels in the sensor. This type of noise is SD and has statistics proportional to the original signal. Therefore, brighter pixels have larger intensities of noise and are assigned with adaptively lower weights in the proposed method, which also demonstrates the effectiveness of our proposed method in achieving robustness to pixel noise. Fig. 12(d) shows the weight assignment for the sparsity constraint of pixels' abundances. It can be observed that the highly mixed regions such as the transition regions of tree and soil are assigned with low weights to weaken the strength of sparsity constraint, while the regions of high sparsity levels are assigned with high weights to promote sparsity accordingly.

Fig. 14 displays the endmember signatures obtained by different methods with comparison to the reference signatures. It can be observed that all the six methods have shown quite similar estimation results and estimated the four target endmembers correctly in general. For the target of water, MVCNMF performs worse than other methods. The endmember signatures extracted by CSsRS-NMF are in good accordance with the reference signatures. Table II shows the SAD results of different methods on the Jasper data. The rows show the results of four targets of interest, i.e., tree, water, soil, and road, along with the mean values, respectively. As shown in the table, CSsRS-NMF achieves the lowest SAD values in three targets, including Tree, Water, and Soil, as well as the mean value. The abundance maps obtained by different methods are presented in Fig. 15. The proposed CSsRS-NMF achieves comparatively satisfactory abundance results. Compared to other methods, the abundance maps generated by CSsRS-NMF are more consistent to the reference maps, especially for the maps of tree and road.

TABLE II
SAD OF ALGORITHMS ON THE JASPER DATA SET

Endmember	Spectral Angle Distance					
	N-FINDR	MVCNMF	$\ell_{2,1}$-NMF	ℓ_1-CENMF	$\ell_{1/2}$-RNMF	CSsRS-NMF
Tree	0.1474	0.1505	0.1728	0.1820	0.1814	**0.1467**
Water	0.0953	0.3156	0.2561	0.2816	0.2930	**0.0623**
Soil	0.1166	0.1076	0.1278	0.1260	0.1345	**0.0788**
Road	0.0906	0.0869	**0.0451**	0.0497	0.0463	0.0486
Mean	0.1125	0.1652	0.1504	0.1598	0.1638	**0.0841**

The second used hyperspectral data in the experiment are the Urban data set collected by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor over an area located at Copperas Cove near Fort Hood, TX, U.S., in October 1995. In this Urban data set, the spectral and spatial resolutions are 10 nm and 2 m respectively. The data set contains a suburban residential area as shown in Fig. 16, with a size of 307×307. There are four kinds of targets existing in this area, including asphalt, grass, roofs, and trees. The data set is composed of 210 spectral bands ranging from 400 to 2500 nm. Due to the water absorption and atmosphere effects, there exists some noisy bands with low SNR values in the Urban data set, including the bands 1–4, 76, 87, 101–111,

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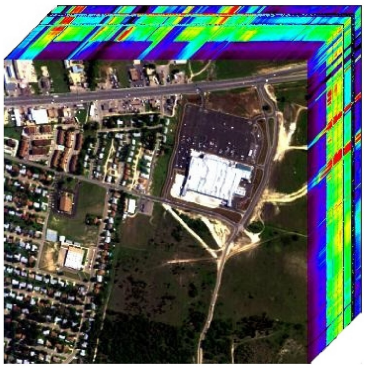


Fig. 16. Hyperspectral Digital Imagery Collection Experiment (HYDICE) Urban Data Set.

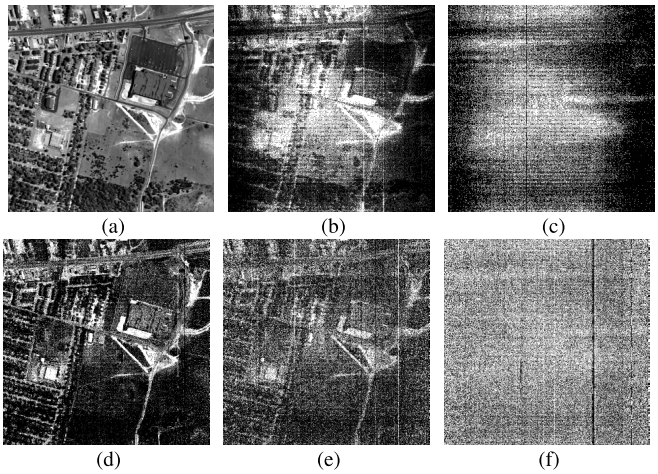


Fig. 17. Images of some bands in the Urban data set. (a) Band 134. (b) Band 139. (c) Band 146. (d) Band 206. (e) Band 207. (f) Band 209.

TABLE III
SAD OF ALGORITHMS ON THE URBAN DATA SET WITH NOISY BANDS REMOVED

Endmember	Spectral Angle Distance					
	N-FINDR	MVCNMF	$\ell_{2,1}$-NMF	ℓ_1-CENMF	$\ell_{1/2}$-RNMF	CSsRS-NMF
Asphalt	0.1930	0.1148	0.1258	0.3252	0.1482	**0.1073**
Grass	1.3748	0.3952	0.1964	**0.1148**	0.1676	0.2441
Tree	0.0848	0.1527	0.0743	0.1200	0.1031	**0.0304**
Roof	**0.1676**	0.5396	0.8273	0.6478	0.7585	0.3026
Mean	0.4550	0.3006	0.3060	0.3020	0.2943	**0.1711**

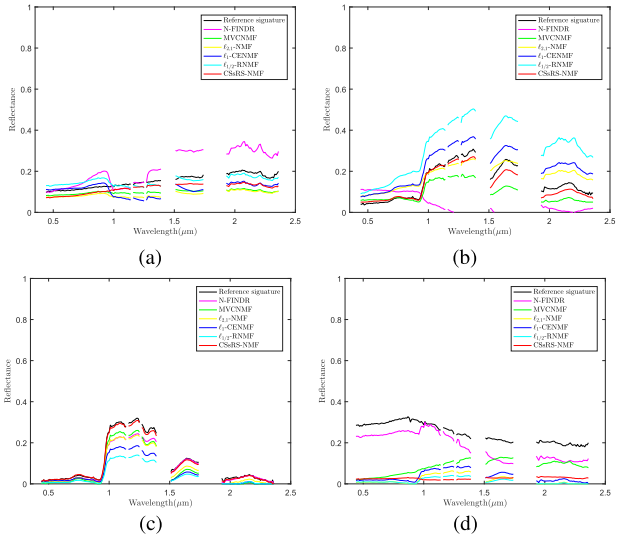


Fig. 18. Comparison of endmember signatures estimated by different methods over the Urban data without noisy bands. (a) Asphalt. (b) Grass. (c) Tree. (c) Roof.

136-153, and 198–210. Fig. 17 displays some band images of the Urban data set. As shown in the figure, band 134 is a low-noisy band, while bands 139, 146, 206, 207, 209 are noisy bands corrupted by noise in varying degrees. In addition, the hyperspectral data also suffer from the corruption of noisy pixels such as stripes, which can be observed in Fig. 17(b)–(f). In the traditional unmixing methods, the bands with low SNR values are usually manually removed and the noisy pixels are not considered separately. The existence of noisy bands and noisy pixels seriously degrades the performance of most existing unmixing methods. However, in real-world applications, it may not be practical to eliminate the noisy bands or noisy pixels directly in the unmixing process since the prior knowledge of the noisy bands or noisy pixels may not be available and the noisy part may also contribute important information and should not be discarded. In the experiments, we examine the performance of algorithms on data removing noisy band images (i.e., $L = 162$) and data containing noisy band images (i.e., $L = 210$) to explore the influence of noisy part in the data. In addition, since the ground truth of abundances is unavailable, experiments conducted on both data without noisy bands and data with the noisy bands can be used to evaluate the unmixing performance of algorithms on abundances effectively through comparing the abundance maps of the two experiments.

1) Results without the low-SNR band images: We first present the unmixing results of algorithms on the Urban data set with the low-SNR band images removed. Table III represents the SAD results of different methods. The rows show the results of four targets of interest, i.e., asphalt, grass, tree, and roof, along with the mean values, respectively. As shown in Table III, it can be seen that CSsRS-NMF achieves SAD results better than other methods in general. The proposed CSsRS-NMF achieves the best results in two targets, including asphalt and tree, as well as the mean value. Fig. 18 displays the endmember signatures obtained by different methods with comparison to the reference signatures, which demonstrates the endmembers extracted by CSsRS-NMF are in good accordance with the reference signatures. It can be observed that apart from the proposed method, ℓ_1-CENMF and $\ell_{1/2}$-RNMF generally achieve better results than the others. However, in general, CSsRS-NMF obtains the best performance. Fig. 19 presents the abundance maps obtained by different methods on the Urban data set with noisy bands removed. As can be observed from Fig. 19, the abundance maps achieved by CSsRS-NMF accord well with the reference maps.

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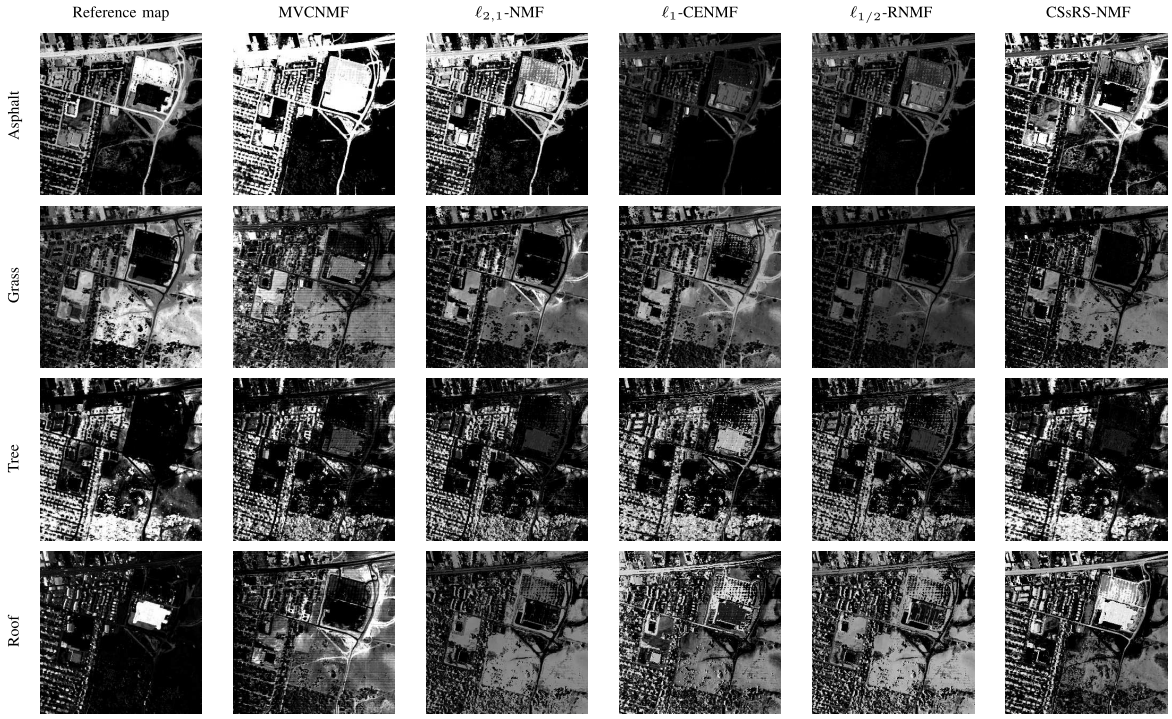


Fig. 19. Abundance maps of the Urban data without noisy bands by MVCNMF, $\ell_{2,1}$-NMF, ℓ_1-CENMF, $\ell_{1/2}$-RNMF, CSsRS-NMF, along with the reference maps. Each row shows the corresponding abundance maps of the same endmember.

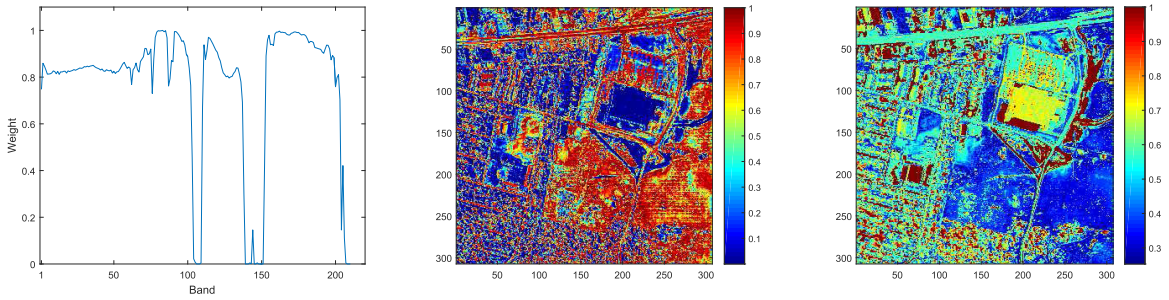


Fig. 20. Weight assignment of band, pixel, and sparsity constraint when applying CSsRS-NMF on the Urban data set containing noisy bands. (a) Weight for each band. (b) Weight for each pixel. (c) Weight for sparsity constraint of each pixel's abundances.

2) Results with the low-SNR band images: We also test the algorithms on the Urban data set containing the noisy bands, which includes all the 210 bands. Fig. 20 presents the weight assignment of CSsRS-NMF in the last iteration when applied to the Urban data set containing the noisy bands. Fig. 20(a) shows that CSsRS-NMF can assign adaptive weights to different bands with various noise levels. The noisy bands are assigned with low weights according to their noise levels and the low-noisy bands are assigned with relatively high weights, which is in good agreement with the observation in Fig. 17. Fig. 20(b) shows the weights assigned for pixels by CSsRS-NMF. The pixels have low reconstruction errors are assigned with higher weights and the contribution of noisy pixels is suppressed, which helps to make better use of pixels' information. It can also be observed that the regions with brighter pixels such as the roof and asphalt areas are assigned with lower weights. In these areas, the pixels have high intensities of signal and carry noise dominated by SD noise. Therefore, the reconstruction errors of these high-noisy pixels are relatively high and the corresponding weights assigned for these pixels by CSsRS-NMF are low.

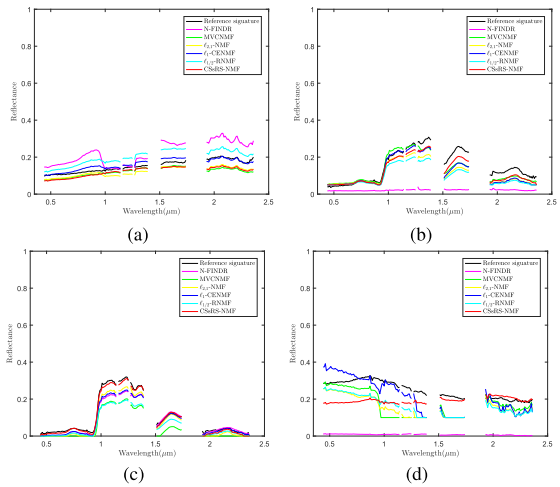


Fig. 21. Comparison of endmember signatures estimated by different methods over the Urban data containing noisy bands. (a) Asphalt. (b) Grass. (c) Tree. (c) Roof.

This phenomenon validates the robustness of CSsRS-NMF to pixel noise. Fig. 20(c) shows the weight assignment for the sparsity constraint of pixels' abundances. The highly mixed

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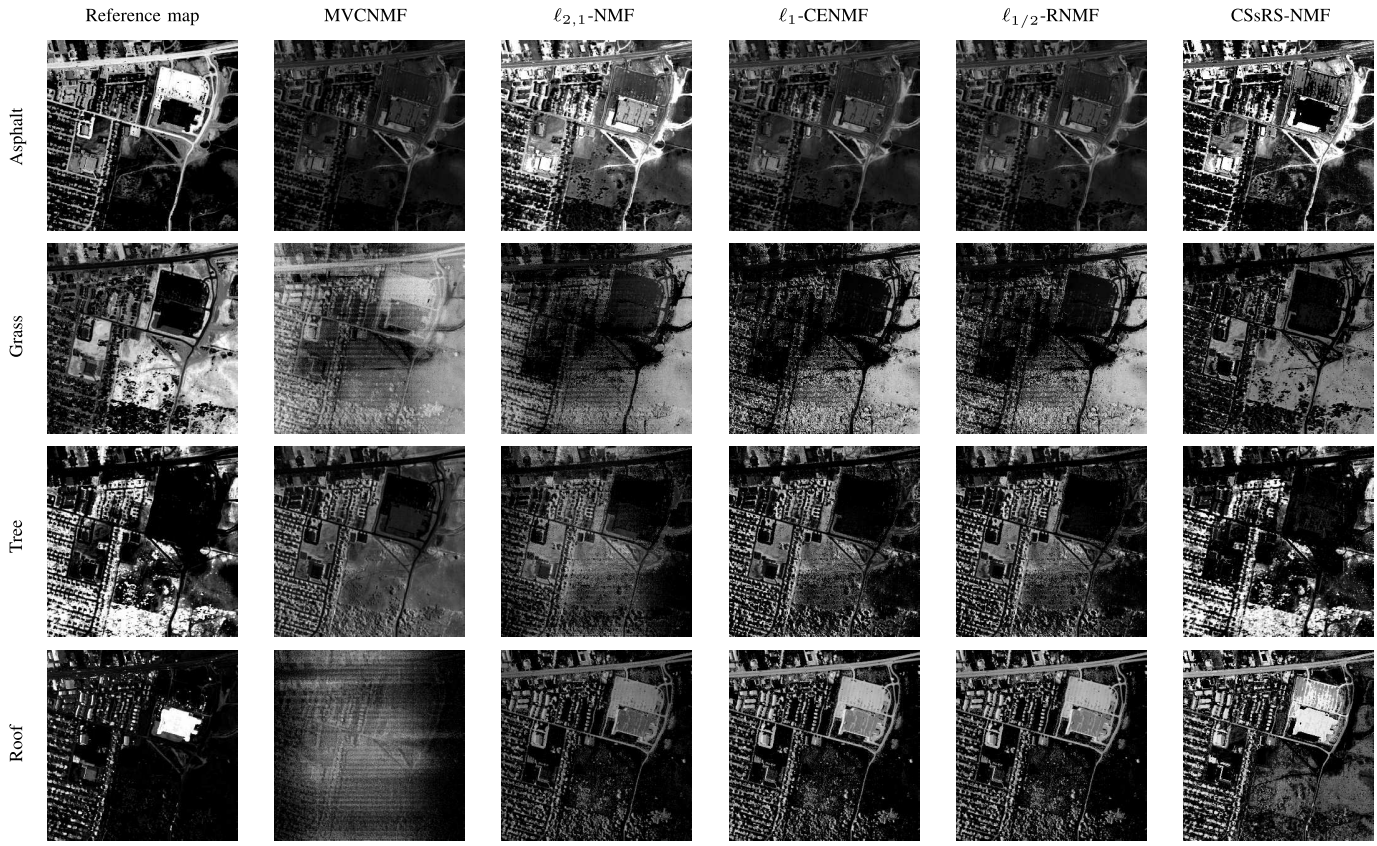


Fig. 22. Abundance maps of the Urban data containing noisy bands by MVCNMF, $\ell_{2,1}$-NMF, ℓ_1-CENMF, $\ell_{1/2}$-RNMF, CSsRS-NMF, along with the reference maps. Each row shows the corresponding abundance maps of the same endmember.

regions such as the transition regions of tree and grass are assigned with low weights to weaken the strength of sparsity constraint, while the regions of high sparsity levels such as the roof regions are assigned with high weights to promote sparsity accordingly. Table IV shows the SAD results of different methods. CSsRS-NMF also achieves SAD results better than other methods in general. The proposed CSsRS-NMF achieves the best results in two targets, including grass and tree, as well as the mean value. Fig. 21 displays the endmember signatures obtained by different methods and the reference signatures for visual comparison, which demonstrates the endmembers extracted by CSsRS-NMF are in good accordance with the reference signatures in general. Comparing with other methods, CSsRS-NMF generates a good estimate on the endmembers of asphalt, grass, and tree but underestimates the endmembers of the roof. Fig. 19 presents the abundance maps obtained by different methods on the Urban data set containing noisy bands. Comparing Figs. 19 and 22, we can investigate the influence of noisy bands and evaluate the performance of algorithms on abundances. Apparently, the abundance maps of MVCNMF are seriously corrupted by the noisy bands. The other three compared algorithms show some robustness as designed. But the abundance results of these methods are still inevitably corrupted by the noisy band images. For instance, the abundance maps of tree achieved by $\ell_{2,1}$-NMF [32], ℓ_1-CENMF [25], and $\ell_{1/2}$-RNMF [24] are all influenced by the stripe noise existing in the noise band images. For CSsRS-NMF, the abundance maps achieved on data set containing the noisy bands are quite similar to the results achieved without noisy bands and are also in good agreement with the reference maps, which effectively demonstrates that the proposed CSsRS-NMF has good robustness to noise and can impose adaptive weighted sparsity constraint to improve unmixing performance.

TABLE IV
SAD OF ALGORITHMS ON THE URBAN DATA SET CONTAINING NOISY BANDS

Endmember	Spectral Angle Distance					
	N-FINDR	MVCNMF	$\ell_{2,1}$-NMF	ℓ_1-CENMF	$\ell_{1/2}$-RNMF	CSsRS-NMF
Asphalt	0.1422	0.1333	0.0932	**0.0840**	0.0856	0.1148
Grass	0.4227	0.3422	0.3746	0.3983	0.4098	**0.2451**
Tree	0.1311	0.2021	0.1298	0.0900	0.0962	**0.0296**
Roof	**0.2579**	0.6882	0.6282	0.5101	0.5396	0.3116
Mean	0.2385	0.3414	0.3065	0.2706	0.2828	**0.1753**

V. CONCLUSION

In this article, we propose a correntropy-based spatial-spectral robust sparsity-regularized unmixing model to achieve 2-D robustness and adaptive weighted sparsity constraint for abundances. By introducing the HQ technique, the robust sparsity-regularized unmixing model is optimized and the updating rules of the proposed model are provided. Compared to the existing methods, the proposed CSsRS-NMF

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firstly considers the noise of hyperspectral data in both spatial dimension and spectral dimension and incorporates correntropy-based adaptive sparsity constraint, which achieves robustness to both band noise and pixel noise and enforces sparsity constraint with weight adaptive to sparsity level of abundance. The experimental results on the synthetic data and real HSIs show that the proposed CSsRS-NMF achieves much better unmixing performance compared to the state-of-the-art methods.

For the future work, the performance of our method could be further improved by incorporating various other regularizers. Through the HQ technique, various potential functions φ_R and φ_S could be explored to develop more robust sparse unmixing methods accommodating the needs for different applications. More methods based on mining and making full use of information latent in hyperspectral data would also be worthy of further study.

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